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A new transferable deep learning approach for crop mapping

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ABSTRACT

Accurate crop mapping provides important information for government decision-making and agricultural management. Recent advancements in deep learning have significantly enhanced the capabilities of remote sensing-based crop mapping. In this study, we developed a new deep learning approach, DSH, which comprises three modules, DeepLabV3+ (D), channel self-attention (S), and histogram matching (H). The DeepLabV3+ module learns the spatial distribution of crops in the spatial and spectral dimensions. The channel self-attention module was used to enhance the weights of the important features. The histogram-matching module addresses the domain gap between the training and testing images and enhances the transferability of the DSH. In addition, the input data of the DSH are monthly synthesized Sentinel-2 data, thus relaxing the data collection requirements. The performance of the DSH was evaluated and benchmarked against HRNetV2, DeepLabV3+, and random forest (RF) at eight sites in the U.S. China, and France with different farming structures, climate conditions, and landscape complexities. Temporal transfer revealed that the classification performance of DSH improved as the growth season progressed, peaked in the peak growth season (August), and then declined at each study site. The DSH model trained during the peak growth season demonstrated superior temporal transferability. Spatial transfer experiments demonstrated that transferring DSH to sites with similar planting structures achieved higher precision than full spatial transfers. Spatiotemporal transfer experiments conducted at eight sites over three years (2020–2022) demonstrated an average overall accuracy (OA) of 88.1% for DSH, highlighting the importance of similar planting structures for effective spatiotemporal transfer. DSH outperformed HRNetV2, DeepLabV3+, and RF in OA, producer's accuracy, and user's accuracy at all eight study sites. Notably, DSH exhibited comparable performance during both the peak and full growth seasons, with an average OA of 93.9%. The classification accuracy of test images after histogram matching remained stable from 2020 to 2022, whereas the accuracy of raw test images fluctuated annually. This indicates that histogram matching is effective in improving the transferability of DSH. Embedding the channel self-attention module into the position of the high-level features had a more significant impact on improving DSH's performance than other configurations. Visualizing features from the channel self-attention and high-level layers revealed that integrating the channel self-attention module with DeepLabV3+'s high-level features significantly improved crop mapping by enhancing relevant semantic information. Overall, DSH exhibits robust spatiotemporal generalizability and requires minimal input data, making it suitable for large-scale crop mapping.

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Sentinel-2 synthesized imagery; deep learning; channel self-attention; histogram matching; spatiotemporal transferability

1. Introduction

Accurate and timely crop mapping provides important information for government decision-making and various agricultural applications, including grain supply chain management, crop insurance, and disaster risk management (Lobell et al. 2015; Weiss, Jacob, and Duveiller 2020). Therefore, the development of a generalizable crop mapping approach is urgently required. In recent years, crop mapping methods based on deep learning (DL) have emerged as a prominent research focus in the field of remote sensing (Benami et al. 2021; Castro, Queiroz Feitosa, and Nigri Happ 2018; Rußwurm and Körner 2020).

Crop mapping methods primarily encompass two types: traditional Machine Learning (ML) methods and DL methods. ML methods, such as support vector machines (Mountrakis, Im, and Ogole 2011), decision trees (Song et al. 2017), and random forest (RF) (Feng et al. 2019; Hao et al. 2020; Pelletier et al. 2016; You et al. 2021) have been widely applied. However, their primary drawback is a heavy reliance on manual

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feature engineering based on domain knowledge, coupled with high learning costs for model transferability (Li et al. 2023). For instance, time series data of synthetic aperture radar (SAR) backscatter coefficients, such as vertical transmit/vertical receive (VV), vertical transmit/horizontal receive (VH), and vegetation indices (VIs) including normalized difference vegetation index (NDVI), normalized difference yellow index (NDYI), Red-edge position (REP), normalized difference red edge (NDRE) have been specifically utilized for mapping crops like rapeseed, maize, and rice (Jiang et al. 2024; Qi et al. 2024; You et al. 2023; Zang et al. 2023). Additionally, VIs have been employed to predict corn yields in the U.S. corn belt (Li et al. 2024; Ma et al. 2021). Despite moderate success, it remains challenging for predefined features to be discriminative for different crops across large and heterogeneous areas, and model's transferability is often suboptimal. Moreover, most ML methods can only extract low- or middle-level features from the original data and perform poorly in extracting contextual spatial information (Zhang, Pan, et al. 2020). Recent advances have shown that DL methods can learn essential spatial dependencies from remote sensing observations. Compared to ML methods, DL models, with their deeper and more complex structures, can automatically learn useful information from a multitude of features at various levels of representation, achieving high classification accuracy. Convolutional neural networks (CNNs), a classical DL method, have been successfully employed for crop mapping using models such as HRNet (Goyal et al. 2023), DeepLabV3+ (Zhang, Ding, et al. 2020), U-Net (Ge et al. 2021; Jo et al. 2023), among others (Kussul et al. 2017; Zhong, Hu, and Zhou 2019, Zhong et al. 2019).

The DeepLabV3+ network, proposed by Google in 2018 (Chen et al. 2018), is a CNN that includes an Atrous Spatial Pyramid Pooling (ASPP) module. The ASPP contains multiple parallel convolution branches with different dilated convolution rates for multi-scale feature extraction. Dilated convolution enables DeepLabV3+ to achieve a larger receptive field, which is essential for capturing the spatial distribution of scattered or clustered crop plots. Furthermore, multi-scale feature extraction enhances DeepLabV3+'s capacity to acquire richer contextual information. Since crop plots often represent targets at various scales in remote sensing imagery, the DeepLabV3+ network structure possesses inherent advantages for remote sensing crop mapping. Like many encoder-decoder CNNs, DeepLabV3+ captures complex spatial context relationships by expanding the dimensions of its feature representations. To reduce the number of parameters and computational complexity that typically results from such dimensional expansion, DeepLabV3+ employs weight sharing across different feature levels. However, DeepLabV3+ does not explicitly consider the correlation between those high-level features, which may limit its performance in certain scenarios. In 2017, self-attention mechanisms were proposed for modeling global relationships. Transformer models have become popular for sequence prediction in the field of natural language processing (Vaswani et al. 2017). Recently, some studies have integrated the attention mechanism to improve the performance of CNNs in various classification tasks (He et al. 2022; Li et al. 2020; Wang et al. 2017), and few studies have explored this integration in crop mapping (Rußwurm and Körner 2020; Wang et al. 2021). Unlike traditional convolution operations, which share weights across layers and potentially average relationships between features, the self-attention mechanism allows for dynamic weighting of features based on contextual correlation. This enriches the model's feature representation and improves its generalization. Further research is needed to fully understand the role of self-attention mechanisms in capturing correlations between high-level features in DeepLabV3+.

In large regions, due to variations in crop varieties, terrain, soil conditions, climatic conditions, and agricultural practices, the same crop may exhibit different phenological periods on the same day of the year in different regions, resulting in varying spectral responses (Liu et al. 2019; Lobell et al. 2015). Classifiers trained for a specific-year and specific-region may perform poorly in other years and regions due to changes in spectral reflectance between training and test images. Consequently, developing spatiotemporally transferable crop mapping models remains a significant scientific challenge. Two primary strategies have been proposed to address the spectral distribution variability problem between test and training images. The first strategy focuses on modifying network architectures or loss functions to enhance the transferability of a model in a specific

area (Diakogiannis et al. 2020; Kemker, Salvaggio, and Kanan 2018; Zhang et al. 2019). However, this strategy requires a substantial amount of computing resources to perform additional optimization steps to retrain the entire network. The second strategy enhances the transferability of a model by increasing the training sample size. This involved using training samples from multiple years (Xu et al. 2020) or incorporating a small number of samples from spatiotemporal transfer regions (Hamrouni et al. 2021). However, obtaining a substantial number of samples from multiple years or regions is often challenging. Color equalization algorithms can adjust spectral distribution variability, maintaining a relatively consistent spectral value range (Cao et al. 2015). The histogram matching method, based on statistical principles, can enhance image contrast and correct for the nonlinear response of detectors. To improve the generalization of DL methods for crop mapping under the challenge of phenological period differences, further testing of the impact of the histogram matching method on classifier models is necessary.

To date, most DL-based crop mapping studies have utilized multi temporal remote sensing data throughout the full growth season (Blickensdörfer et al. 2022; Cai et al. 2018; Jinfan et al. 2021; You et al. 2021; Zhong, Hu, and Zhou 2019), with few studies employing single temporal remote sensing data (Mathur and Foody 2008; Yang, Everitt, and Murden 2011). Ge et al. (2021) trained a deep learning model using data synthesized from median values during the sowing and peak growth seasons and achieved satisfactory accuracy in spatial transfer testing. Sun et al. (2020) identified multiple crops in the North Dakota region using remote-sensing data from the silking, milking, and ripening stages, achieving a level of accuracy comparable to that of the United States Department of Agriculture Cropland Data Layer. Xu et al. (2020) used a Long Short-Term Memory (LSTM)-based model to identify maize and soybeans using time-series Landsat data. The results showed that, although the model exhibited a certain spatiotemporal transferability, its performance significantly decreased when data was missing. The crop mapping method based on multi temporal data can only be executed after acquiring the necessary images, which increases the complexity of the data acquisition process due to the impact of clouds and shadows. Moreover, even when sufficient multi temporal data are obtained through multi source remote sensing, additional data fusion and preprocessing remain time consuming and labor intensive (Gao et al. 2017; Waldhoff, Lussem, and Bareth 2017). In contrast, crop mapping methods based on single-phase images or synthesized images face fewer constraints regarding data availability.

We propose a DL model with a high level of spatiotemporal transferability that is suitable for crop mapping, named as DSH. This model is based on DeepLabV3+ (D), enhances the weights of important features through the self-attention mechanisms (S), and addresses domain differences between the training and test images through histogram matching (H). The model's input is monthly synthesized remote sensing data, reducing the input data requirements. The performance of DSH has been tested at eight sites with different farming structures, geolocations, and landscapes in the U.S., China, and France with five specific questions: 1) How does the temporal transferability of DSH change with crop growth? 2) How does the spatial transferability of DSH differ between full spatial transfer and transfer under similar planting structures? 3) How does the spatiotemporal transferability of DSH vary across different regions? 4) How does the classification performance of the DSH model compared with HRNetV2, DeepLabV3+, and RF? and 5) What are the performance differences between the DSH model trained using peak growth season images and the model trained using the full growth season images?

2. Study sites and data

2.1. Study sites

Eight study sites with different farming structures, geolocations, and landscapes were selected for testing the algorithm, and the choice of these sites was oriented toward having a range of major crops (Figure 1). Sites 1–3 are situated in the central United States, with each site covering 55 km × 55 km (Figure 1(a)). Site 1 is situated in Minnesota ($89°34' - 97°12'$W and $43°34' - 49°23'$N). Site 2 is situated in Nebraska ($95°25' - 104°$W and $40°–43°$N). Maize and soybeans were the dominant crop types at sites 1 and 2 (Figure 2). Site 3 is situated in Arkansas ($89°41' - 94°42'$W and $33°–36°30'$N). Maize, rice, and soybeans were the main grain crops at site 3, but the croplands at site 3 were mixed with woody wetlands

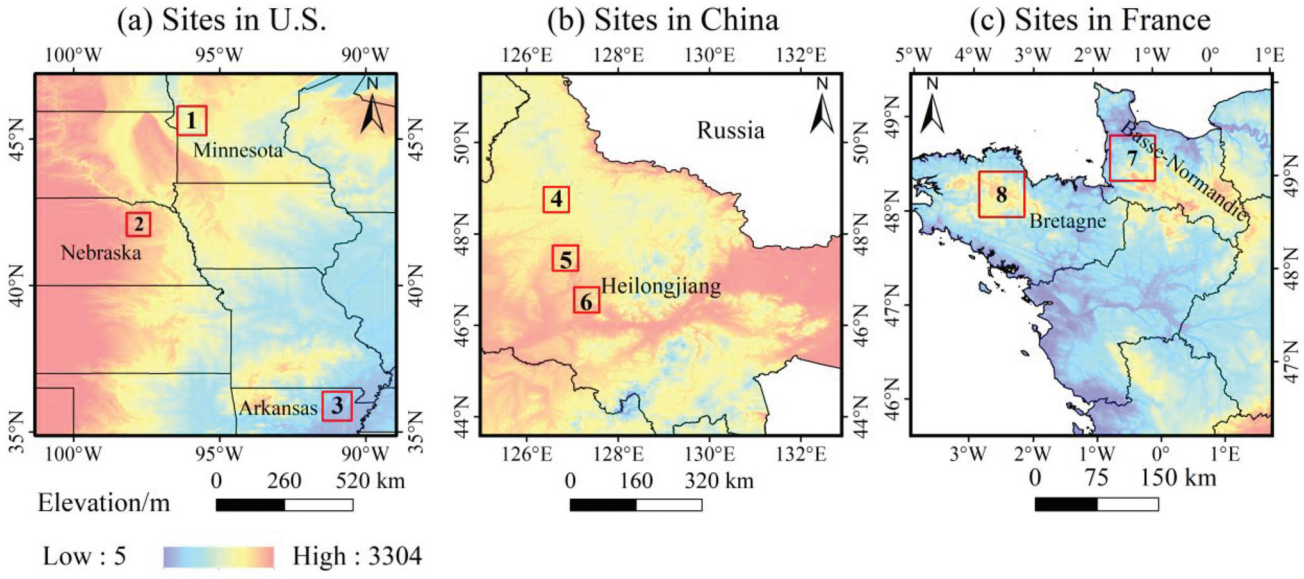

Figure 1. Location of study sites in the United States (a), China (b), and France (c). Digital elevation Model (DEM) is displayed.

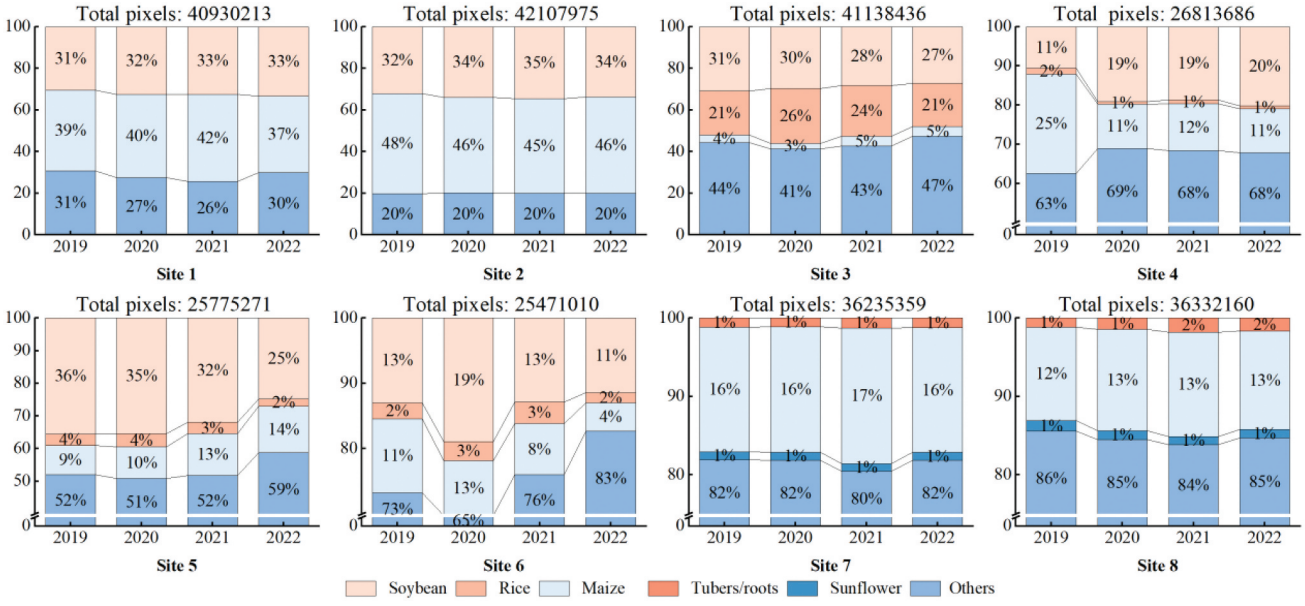

Figure 2. The total sample size as well as the proportion of samples for specific crop cover types in each site and each year.

and forests. Sites 4–6 are situated in Heilongjiang Province (HLJ) in China, with each site covering 35 km × 35 km (Figure 1(b)). HLJ is situated in the northeast of China (121° – 125°E and 43° – 53°N) and is the largest grain-producing province in China. Maize, rice, and soybeans were the main crop types at sites 4–6. Sites 7–8 are situated in the northwest France, with each site covering 55 km × 55 km (Figure 1(c)). Site 7 is situated in the Basse-Normandie region in France (2°3′W – 0°58′E and 48°10′ – 49°43′N). Site 8 is situated in Bretagne region (5°8′–1°1′W and 47°16′–49°6′N). Basse-Normandie and Bretagne are significant agricultural regions in France. Maize was the predominant crop at sites 7–8, with small amounts of sunflower and tubers/roots also being cultivated at these two sites.

All these sites were planted with single-season crops. The maize growth season extends from mid-May to late September, while the soybean growth season spans from late May to late September. Rice, tubers/roots, and sunflower are typically planted from April to early June, reaching maturity in late

September. These crops share similar phenological periods, which poses a challenge to crop mapping by remote sensing. The average area of farmland at sites 1–4 was relatively large, whereas that at sites 5–8 was relatively fragmented. Among the eight sites, sites 1–2 had the same main crops, maize and soybean; sites 7–8 had maize as the main crop; and sites 3–6 had maize, rice, and soybean as the main crops, but with very different planting proportions. Site 3 mainly planted rice and soybean, while sites 4–6 mainly planted maize and soybean. We selected sites 1–2 and 7–8 to compare the spatiotemporal transfer of models with similar planting structures, and sites 3–6 to compare the spatiotemporal transferability of models with different planting structures.

2.2. Data and preprocessing

Experiments were conducted using Sentinel-2 Level-2A surface reflectance (SR) data. The orbit and scene numbers of the Sentinel-2 images used in this study are listed in Table 1. Ten optical bands, that is, Blue (B), Green (G), Red (R), Red Edge 1 (RE 1), Red Edge 2 (RE 2), Red Edge 3 (RE 3), Near-infrared (NIR), Red Edge 4 (RE 4), SWIR 1, and SWIR 2, were selected for crop mapping. We first synthesized a high-quality monthly Sentinel-2 SR dataset using the quality Mosaic function with the normalized difference vegetation index (NDVI) as a quality control band during the growth season from May 1 to October 30 for the years 2019–2022 (Roy et al. 2010). To ensure spatial resolution consistency, each band was resampled to a 10 m resolution. Coastal aerosols, water vapor, and cirrus bands with 60 m resolution were removed. All the aforementioned processes were carried out on the Google Earth Engine cloud computing platform (Gorelick et al. 2017).

Table 1. The orbit and scene numbers in each site and each year of the sentinel-2 images used in this study.

		Scenes number			
Sites	Orbit number	2019	2020	2021	2022
Site 1	T14TQR, T14TPR,	110	113	110	111
Site 2	T14TNL, T14TNM, T14TPL, T14TPM	148	150	147	146
Site 3	T15SYV, T15SXV	148	144	146	146
Site 4	T51UYP, T51UYQ, T52UCU, T52UCV	292	288	292	294
Site 5	T51TYN, T52TCT	144	144	146	148
Site 6	T52TCS	74	77	73	74
Site 7	T30UWV, T30UXU, T30UXV	185	185	186	193
Site 8	T30UVU, T30UWU	146	146	147	146

Datasets of the crops at sites 1–3 were collected from the United States Department of Agriculture (USDA) Cropland Data Layer (CDL) (Boryan et al. 2011; Han et al. 2012). Datasets of the crops at sites 4–6 were collected from the National Bureau of Statistics of China. Datasets of the crops at sites 7–8 were collected from Theia's land cover product (Theia 2019). We categorized the crop types at sites 1–2 as corn, soybean, and "others," at sites 3–6 as corn, soybean, rice, and "others," and at sites 7–8 as sunflower, corn, tuber/root crops, and "others." Here, tuber/root represents beet and potato. The "others" class at each site includes other crops, water bodies, buildings, forest, and meadows.

The total numbers of sample pixels at sites 1–8 were 40,930,213, 42107,975, 41138,436, 26813,686, 25775,271, 25471,010, 36235,359, and 36,332,160, respectively. The proportions of crop types at these eight sites are illustrated in Figure 2. In 2022, the proportion of crop types was above 50% at sites 1–3, above 30% at sites 4–5, 17% at site 6, 18% at site 7, and 16% at site 8. The amount of sample data for different land cover types exhibited a noticeable imbalance. Spatiotemporal transfer across these sites provides an opportunity to evaluate the model performance on the datasets with various unbalanced land cover types.

3. Methodology

3.1. DSH model

The proposed DSH model comprises three modules, namely DeepLabV3+, channel self-attention, histogram matching, and a classifier (Figure 3). The DeepLabV3+ module learns the spatial distribution of crops in the spatial and spectral dimensions. The channel self-attention module is used to enhance the weights of the important features. The histogram matching module is used to adjust the spectral value range of the test image to ensure that it was consistent with the spectral distribution of the training image to enhance the transferability of the model. The classifier projects the feature map to the classification space, obtaining pixel-level classification results.

3.1.1. DeepLabV3+ module

The DeepLabV3+ module consists of an encoder and a decoder. The encoder utilizes the MobileNetV2 as its

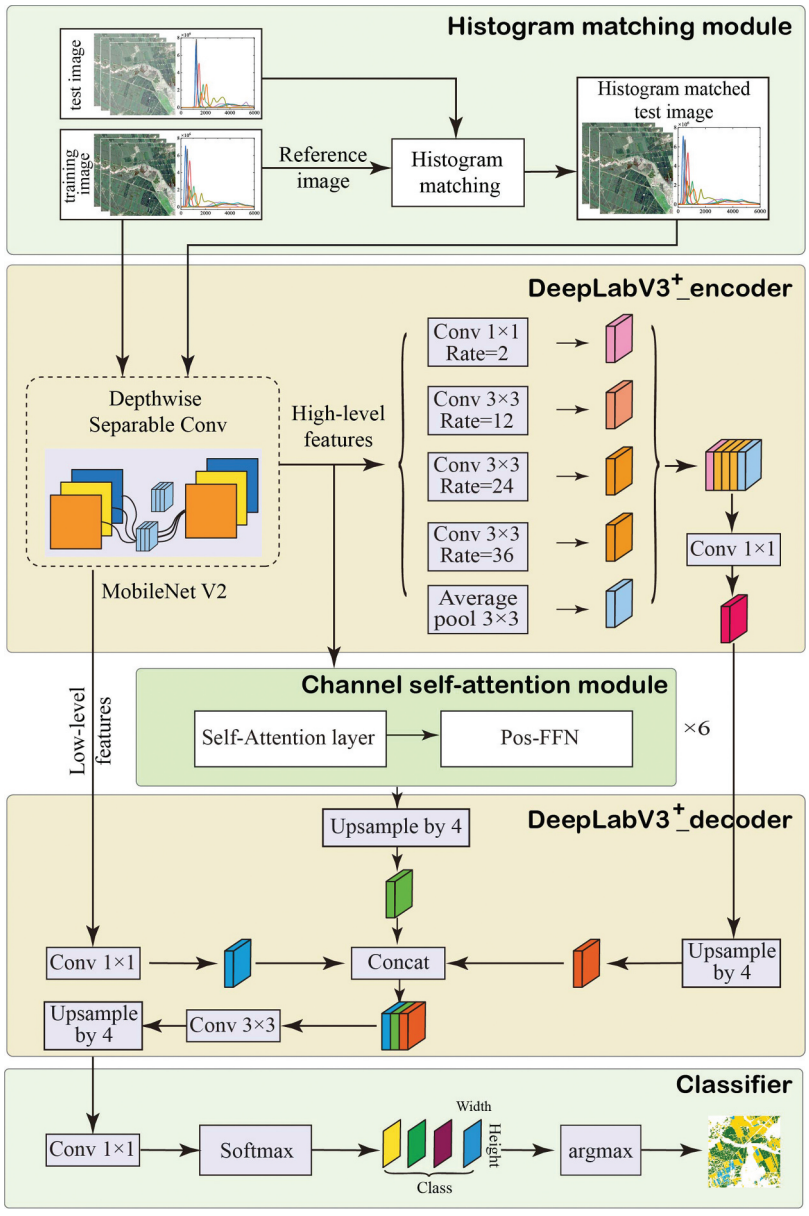

Figure 3. The framework of the DSH for crop mapping: incorporating DeepLabV3+, channel self-attention, histogram matching, and classification. The DeepLabV3+ module captures crop spatial distribution, while the channel self-attention module emphasizes crucial features. The histogram matching module aligns the spectral range of the test image with the spectral distribution of the training image, enhancing the DSH's transferability. Finally, the classifier transforms the feature map into a classification space, delivering pixel-level classification outputs.

backbone, along with an ASPP (Chollet 2017). We have replaced Xception in the original DeepLabV3+ with MobileNetV2 based on several considerations: First, MobileNetV2 is a lightweight network that effectively reduces the number of parameters and computational burden of the model, thus allowing the model to run on resource-constrained devices while maintaining accuracy. Second, MobileNetV2 is an improved network that facilitates better reuse and extraction of features at different scales to provide detailed semantic information for DeepLabV3+. In addition, MobileNetV2 provides a scalable framework that makes it easy to resize models to meet specific resource constraints or accuracy requirements.

In this context, MobileNetV2 is used to extract both low- and high-level features from the input images. Low-level features represent some basic characteristics of the objects, such as edges and colors, and help to locate the boundary details of the crop plots, with small receptive field and high spatial resolution. High-

level features capture the abstract semantic information of the image, such as the shape and structure of crop plots, which are used to differentiate and locate different crop plots. They have a large receptive field and low spatial resolution. Furthermore, the ASPP comprises four parallel convolution layers and an average pool layer, which allows ASPP to extract high-level features across multiple scale from the backbone's output.

In the decoder, the high-level features extracted from the ASPP and the channel self-attention module are upsampled to restore their resolution to match the size of the original input. These features are then fused with low-level features to enhance the extraction of boundaries.

For an input image x_0 of shape $H \times W \times C$, the DeepLabV3+ module can be represented as following:

$$x_{low}, x_{high} = BACKBONE(x_0) \tag{1}$$

$$x_i = ASPPConv\left(x_{high}, k_i, d_i\right), i = 0, 1, 2, 3 \tag{2}$$

$$x_4 = Global\ pooling\left(x_{high}\right) \tag{3}$$

$$x_{aspp} = CONCAT\left(x_i + Upsample(x_4)\right), i = 0, 1, 2, 3 \tag{4}$$

$$x_{fuse} = CONCAT\left(x_{low} + Upsample\left(x_{aspp}\right) + Upsample(x_{atten})\right) \tag{5}$$

Where $ASPPConv\left(x_{high}, k_i, r_i\right)$ represents the dilated convolution operation with kernel k_i and dilated rate d_i for the high-level features output x_{high} of the backbone. When $i = 0$, $k_0 = 1 \times 1$, $d_0 = 1$; When $i = 1, 2, 3$, $k_i = 3 \times 3$, d_i is 12, 24, and 36 respectively. $CONCAT(.)$ is concatenate operation. The attention feature, x_{atten}, for the high (low) features is calculated from the self-attention module.

It is noted that the features generated by ASPP are fused not only with low-level features but also with channel self-attention features, as depicted in Equation (5). To optimize multiple scale feature fusion, the channel self-attention module is strategically embedded at distinct positions, as detailed in Figure 4, with the corresponding results discussed in section 5.2 Specifically, P1 represents the embedding position from the high-level features to ASPP. P2 represents the embedding position from the high-level features to the decoder. P3 represents the embedding position from the low-level features to the decoder.

3.1.2. Channel self-attention module

The original DeepLabV3+ focuses on the boundaries and shapes of the objects in a hierarchical manner. In order to capture the correlations of pixels across channels, we improve the model by embedding channel self-attention modules. The channel self-attention module calculates a channel weight for each hidden feature extracted from the high (low) -level feature map. We employed the encoder from the original Transformer network, which consists of six encoder layers, each with the same structure but different weight parameters (Vaswani et al. 2017). Within each layer, there are two sublayers: a multi-head self-attention layer and a simple position-wise feedforward network (Pos-FFN).

Given the feature map x_0 with $H \times W \times C$ obtained from the backbone, we define the pixels as feature elements and serialize them to a 1D sequence X with a shape of $C \times N$, where N is the number of pixels. Pixels in each channel are projected into a high-dimensional space with d_{model} dimensions to learn the correlation of pixels across different channels.

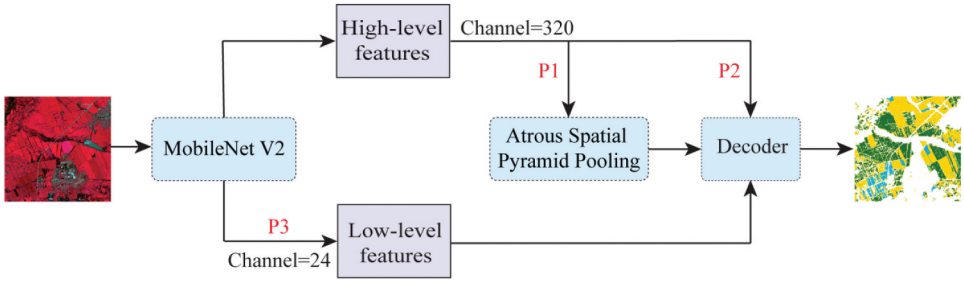

Figure 4. Schematic diagram of different embedding positions of the channel self-attention module. P1 represents the embedding position from the high-level features to ASPP. P2 represents the embedding position from the high-level features to the decoder. P3 represents the embedding position from the low-level features to the decoder. "Channel" denotes the dimension of feature map.

The calculation process of channel self-attention can be represented by the following:

First, *Query*, *Key*, and *Value* are obtained by linear transformations:

$$Query(X) = XW_Q \tag{6}$$

$$Key(X) = X(W_K) \tag{7}$$

$$Value(X) = XW_V \tag{8}$$

Next, the similarity between Query and Key is computed to obtain the attention weights A as follows:

$$A = softmax\left(\frac{Query(X)Key(X)^T}{\sqrt{d_k}}\right) \tag{9}$$

Finally, the output is obtained by weighted sum of Value based on the attention weights:

$$Attention(X) = A\,Value(X) \tag{10}$$

To improve efficiency, a multi-head self-attention layer is utilized, which simultaneously performs the attention function h times.

Where, W_Q, W_K, and W_V are learnable weight matrices, $d_{model} = 2048$, $h = 8$. d_k is the dimension of *Query* or *Key*. For each head of self-attention layer, we made $d_k = d_v = d_{model}/h = 256$.

The output from the multi-head self-attention layer is then transmitted to a position-wise feedforward network, which is a neural network composed of skip connection blocks and normalization operations.

$$\mathcal{F}(x) = (Relu(xW_1 + b_1))W_2 + b_2 \tag{11}$$

$$FFN(x) = Norm(x + \mathcal{F}(x)) \tag{12}$$

Where $\mathcal{F}(x)$ is a two-layer multilayer perceptron (MLP) with a *ReLU* activation function and $Norm(.)$ is the regularization function.

It is noted that the attention features obtained through the channel self-attention mechanism are 1D sequence, and we reformat the x_{atten} with a shape of $H \times W \times C$. So far, we can use the channel self-attention mechanisms to capture pixel-level attention features of low- or high-level features of the backbone.

3.1.3. Histogram matching module

Due to differences in phenological periods, the spectral reflectance values of the same crop are different in images from different regions and years. The domain discrepancy between the training and test images can significantly affect the transferability of classifiers. Specifically, classifiers trained on images from certain regions and years may not perform optimally on images from different regions and years. To address this issue, we employ a histogram matching method designed to align the crop phenological periods in different regions and different years to generate a new set of test images that are closer to the training dataset.

Histogram matching, alternatively termed histogram specification, is an image processing technique that adjusts the distribution of spectral values in one image to match the distribution of another image. The goal of histogram matching is to change the spectral value distribution of the test image so that its statistical characteristics are closer to the histogram of the training image (Gonzalez and Woods 2002).

Before performing histogram matching, it is common to implement histogram equalization on both the training and test images. This is an image processing method that converts the distribution of pixel spectral values in an image to a uniform distribution state, thereby enhancing the image contrast.

The principle of histogram matching is to equalize both training and test image to create the same normalized uniform histograms. Using this uniform histogram as an intermediary, the inverse equalization operation is applied to the test image. This involves applying a specific mapping relationship that converts the pixel values of the test image to a pixel value that matches the training image histogram. The following takes a single band image as an example to illustrate the histogram matching process.

Let the spectral value range of an image is divided into L discrete levels. For the test image, the probability of pixel spectral value r_k (where $k \in [0, L-1]$) is approximated by $p_r(r_k)$. Similarly, for the training image, the probability of pixel spectral value z_m (where $m \in [0, L-1]$ is approximated by $p_z(z_m)$.

Firstly, the test image is histogram equalized, with the equalization transformation T defined as follows:

$$s_k = T(r_k) = (L-1)\sum_{i=0}^{k} p_r(r_i) \tag{13}$$

Likewise, the training image is also histogram equalized, with the equalization transformation G defined as:

$$v_m = G(z_m) = (L-1)\sum_{j=0}^{m} p_z(z_j) \tag{14}$$

Since both images have been equalized, there exists an equivalence relationship: $s_k = \mathrm{T}(r_k) = v_m = \mathrm{G}(z_m)$, that is:

$$(L-1)\sum_{i=0}^{k} p_r(r_i) = (L-1)\sum_{j=0}^{m} p_z(z_j) \tag{15}$$

From Equation (15), it can be seen that to establish a mapping relationship from r_k to z_m, it is only necessary to ensure that the cumulative probabilities of r_k and z_m are as close as possible.

3.1.4. Classifier

In classifier, x_{low}, x_{aspp}, and x_{atten} are fused as x_{fuse} (section 3.1.1). After two convolutional layers and a fully connected layer, a pixel-level prediction class is generated. This process can be represented as follows:

$$x = Conv1 \times 1(Conv3 \times 3(x_{fuse})) \tag{16}$$

$$p = softmax(x) \tag{17}$$

$$\hat{y} = argmax(p) \tag{18}$$

Where p is the pixel-level prediction probability, and $\hat{y}$ is the prediction class.

Considering the imbalanced of the training dataset, we have employed two types of loss functions, namely the cross-entropy and the focal loss function.

The cross-entropy loss function is commonly used in classification with balanced datasets, and it measures the difference between the predicted results and the true labels, as follows:

$$CE = -\sum(y_i * \log(p_i)) \tag{19}$$

Where y_i is the true label, and p_i is the predicted probability of class i.

Focal loss (Lin et al. 2017) can effectively alleviate the impact of class imbalance on model training, as follows:

$$FL = -\sum(a_i * (1-p_i)^{\gamma} * \log(p_i)) \tag{20}$$

Where a_i is the adjustment factor for the class i, and γ is a parameter used to adjust the weights of the samples.

In this study, cross-entropy loss and focal loss function are used for model training at sites 1–2 and 3–8, respectively.

3.2. Baseline models

We conducted a performance comparison between the DSH model and three baseline methods: HRNetV2, DeepLabV3+, and RF. HRNetV2 and DeepLabV3+ are advanced DL methods. HRNetV2 (Sun et al. 2019) sustains high-resolution representations throughout the feature extraction process, thereby enabling the capture of more boundary information. Meanwhile, HRNetV2 processes inputs of different resolutions in parallel, which allows for a more efficient utilization of multi-scale information by the model. Additionally, by fusing features from various resolutions, HRNetV2 is capable of learning a more enriched and comprehensive feature representation. This unique design allows HRNetV2 to obtain richer semantic information and more precise spatial extent. However, the maintenance of high-resolution feature maps throughout the process implies a greater computational burden, particularly with high-resolution inputs, which can lead to relatively slower training and inference speeds for the model. DeepLabV3+ utilizes dilated convolution to effectively increase the receptive field of the network, allowing it to better understand the spatial context within images (Chen et al. 2018). RF is an ensemble of decision trees that provides a strong baseline for classification tasks, particularly in scenarios where interpretability and simplicity are valued (Breiman 2001). Each of these models offers unique insights into the strengths and limitations of different approaches to the crop mapping problem, allowing us to thoroughly evaluate the performance of DSH model.

3.3. Model training

The training sample dataset was generated from the crop layer dataset and remote sensing images from 2019. For each study site, all samples were cropped into 256×256 pixels images blocks with a 50% overlap rate, and image blocks with crop areas of less than 10% were removed to ensure that the training dataset contained sufficient crop pixels. Finally, 2,615, 2,537, 2,271, 2,000, 2,046, 2,107, 2,232 and 2168 pairs of sample image blocks were cropped at study sites 1–8, respectively. For each site, 90% of the image blocks were randomly selected for training purposes, while the remaining 10% were reserved for validation.

The models were initialized using a Gaussian distribution function. The hyperparameters of momentum and epsilon in the batch normalization function were set to 0.9 and 1×10^{-5}, respectively. A stochastic gradient descent (SGD) algorithm (Bottou 2012) with a cosine decay learning rate schedule was selected to optimize the HRNetV2, DeepLabV3+, and DSH models, and the initial learning rate and weight decay were set to 0.001 and 0.0001, respectively. The batch size and epochs were set to 64 and 123, respectively. The other settings of HRNetV2, DeepLabV3+ and DSH are consistent with references (Chen et al. 2018; K. Sun et al. 2019). The RF model was trained in Scikit-learn with the maximum number of features set to 10 and the candidate values for the number of trees set to 100, 200, 500, and 1,000. The classification result with the highest accuracy for different candidate values was identified as the final classification result for RF.

3.4. Experiment design

This study set up the following five sets of experiments to evaluate the classification capability of DSH (Figure 5):

(1) To evaluate changes in temporal transferability of DSH as the crop growth season progressed and to identify the optimal phenological period for model transfer, we trained the DSH using monthly data from the 2019 growth season (base year) for each study site. These models were then applied for crop mapping during the corresponding months in the transfer years of 2020–2022. The growth season, which spans from May to October and includes six months, required the development of six models per study site, each corresponding to a specific month. It is noted that all models were trained only by the data from the base site and base month.

(2) Two categories of experiments were conducted to evaluate the spatial transferability of the DSH model during the peak growth season in 2019. The first experiment involved full spatial transfer, where site 1 was selected as the base site, and sites 2–8 were regarded as the spatial transfer sites. The second experiment focused on spatial transferability among sites with similar planting structures, which was further divided into three groups due to the variability in planting structures across the eight sites. The first group was conducted at study sites 1 and 2, the second group was conducted at sites 3–6, whereas the third group was conducted at sites 7–8. In the first group, the DSH model trained at site 1 (2), was used to classify crops at study site 2 (1). In the second group, the DSH model trained at site 3 (4/5/6) was used to classify crops at sites 4–6 (3,5,6/3,4,6/3-5). The transfer performance of the model at site 3 was measured by averaging the classification performance of the models trained at sites 4–6. A similar method was used to calculate the transfer performance at sites 4–6. In the third group, the DSH model trained at study site 7 (8), was used to classify crops at study site 8 (7).

(3) The model trained during the peak growth season period in 2019 was used to evaluate the spatiotemporal transferability of the DSH model between sites with similar planting structures from 2020 to 2022. The selection of sites and the methodology for calculating accuracy indices were consistent with those of the second type of experiment in set (2).

(4) The performance of the proposed DSH model was benchmarked against HRNetV2, DeepLabV3+, and RF. To ensure a fair comparison, all four models used the same training and testing datasets, which were from each site during the peak growth seasons of 2019 and 2020. The training method for the four models is described in Section 3.3. HRNetV2 and DeepLabV3+ were trained using the same method as for DSH.

(5) To discern the impact of training during peak versus full growth seasons, DSH models trained using monthly synthesized images from the peak growth season and the full growth season (May to October) in 2019 were used to classify crops from 2020 to 2022 at site 2, respectively.

3.5. Accuracy assessment

Model performance was evaluated by comparing the classification results with crop-layer maps. The overall accuracy (OA), producer accuracy (PA), user accuracy (UA), and F1 score (F1) were used to evaluate the crop mapping results (Congalton and Green 2019). The OA, PA, UA, and F1 are formulated as follows:

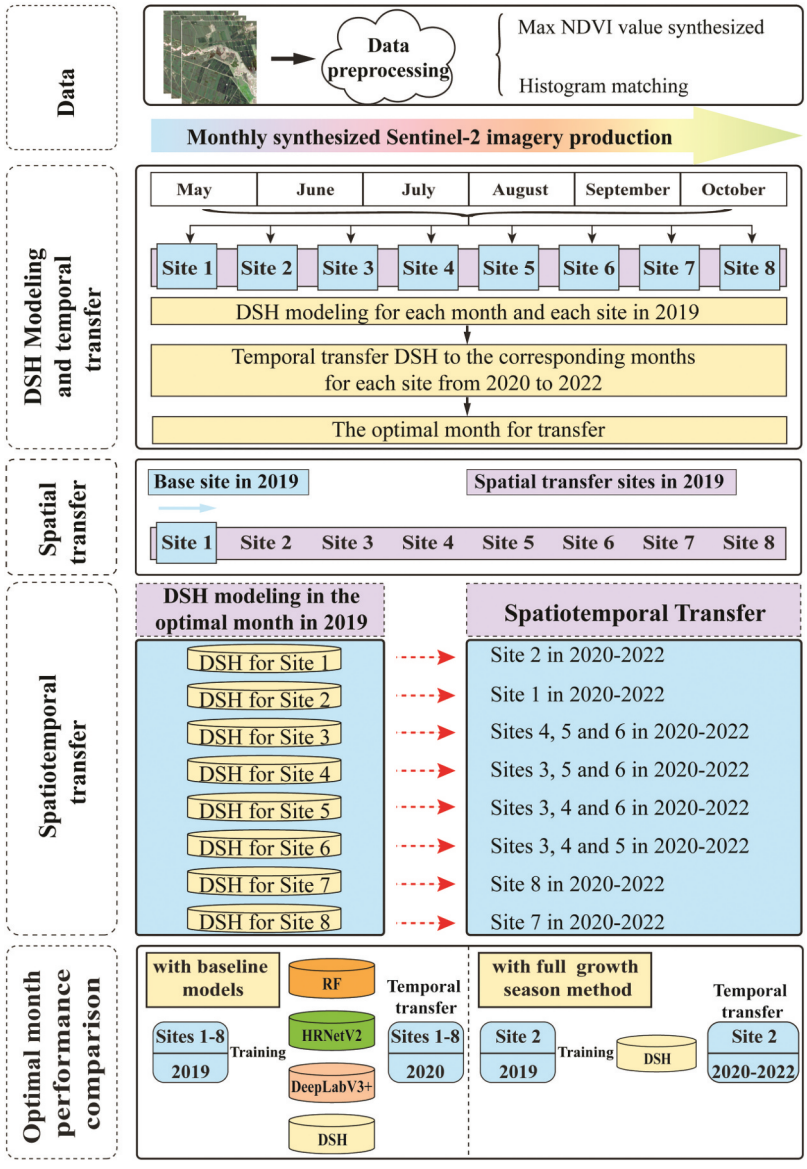

Figure 5. Workflow of the experimental design, which contains five parts: temporal transferability, spatial transferability, spatiotemporal transferability, comparison with baseline models, and comparison with full time series method. It is worth noting that in order to avoid duplication, the second type of experiment of spatial transfer is not listed in this figure because the same sites with similar planting structures were selected as spatiotemporal transfer.

$$OA = \frac{\sum_{i=1}^{C} N_{i,i}}{N} \tag{21}$$

$$PA_i = \frac{N_{i,i}}{\sum_{j=1}^{C} N_{j,i}} \tag{22}$$

$$UA_i = \frac{N_{i,i}}{\sum_{j=1}^{C} N_{i,j}} \tag{23}$$

$$F1_i = 2\frac{PA_i \times UA_i}{PA_i + UA_i} \tag{24}$$

Where C is the total categories of land cover types, N is the total number of validation pixels, $N_{i,i}$ is the number of pixels that are class i and classified into class i, $N_{j,i}$ is the number of pixels that are class j but classified into class i, and $N_{i,j}$ is the number of pixels that are class i but classified into class j. All these metrics range from 0% to 100%.

4. Results

4.1. Assessment of temporal transferability

At each study site, the classification performance of DSH improved as the growth season progressed, peaked in

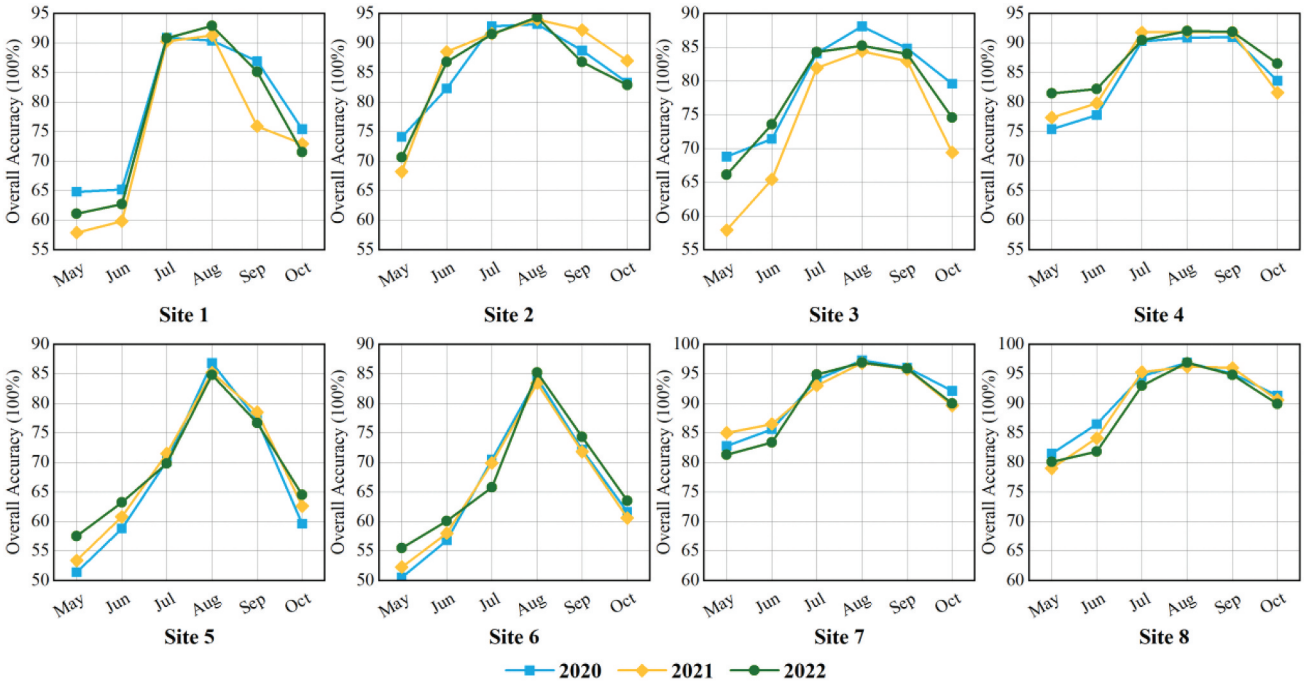

Figure 6. Temporal transferability comparison of month-to-month OA of classification results at eight sites from 2020 to 2022. The start month of the time used for crop mapping is May, e.g. The August in X-axis denotes the time only the composited images of August for crop mapping, exclude the composited images of May, June, and July.

the peak growth season (August), and then declined (Figure 6). In August, the OA exceeded 90% at sites 1, 2, 4, 7, and 8 and reached almost 85% at the other three sites. During the early growth season, the accuracy varied greatly and was relatively low across sites. For example, the OA at most sites did not exceed 70% in May. The significant gap exceeding 35% in OA between August and other months highlights the importance of August in distinguishing the five crops. In August, maize and soybean are in the milk stage, rice is in the pod-setting stages, tubers/roots are in the tuber stage, and sunflower is in the flowering stages. All five crops are in their peak growth season, and the NDVI value is the highest, exhibiting consistent intra-class spectral features and significant inter-class differences. The peak growth season can effectively avoid spectral differences caused by early or late sowing of crops. Meanwhile, the spectral differences caused by different ripening and harvesting times in the late stage of crop growing can also be avoided. Hence, the DSH trained during the peak growth season is more conducive to achieving temporal transferability.

The results of the temporal transferability assessment of the proposed DSH model during the peak growth season are presented in Figure 7. Despite variations in accuracy across sites and crops, the DSH model performed well at all eight sites. Over a span of three years, the model maintained an average OA of 90.9% at eight sites. The average PA of maize, rice, tubers/roots, sunflower and soybean was 86.3%, 85.1%, 62.9%, 55.5% and 92.6%, respectively, at the eight sites from 2020 to 2022, respectively. The average UA of maize, rice, tubers/roots, sunflower and soybean was 81.8%, 80.4%, 62.5%, 61% and 87.5%, respectively. Notably, soybeans performed more successfully than other crops in terms of both the average PA and UA, likely attributed to the larger training sample size at their respective study sites. Conversely, sunflower and tubers/roots experienced higher commission and omission errors, stemming from a training sample size of less than 1% (as shown in Figure 2). Among the sites, site 7 achieved the highest average OA of 97%, while site 6 achieved the lowest average OA of 84.3%. It is significant to mention that the low PA and UA of sunflower and tubers/roots at sites 7 and 8 had no effect on the OA, due to their minimal planting proportion. Figure 8 demonstrates that, in both the regular (sites 1–4) and fragmented (sites 5–8) landscapes, the DSH model achieved good classification results.

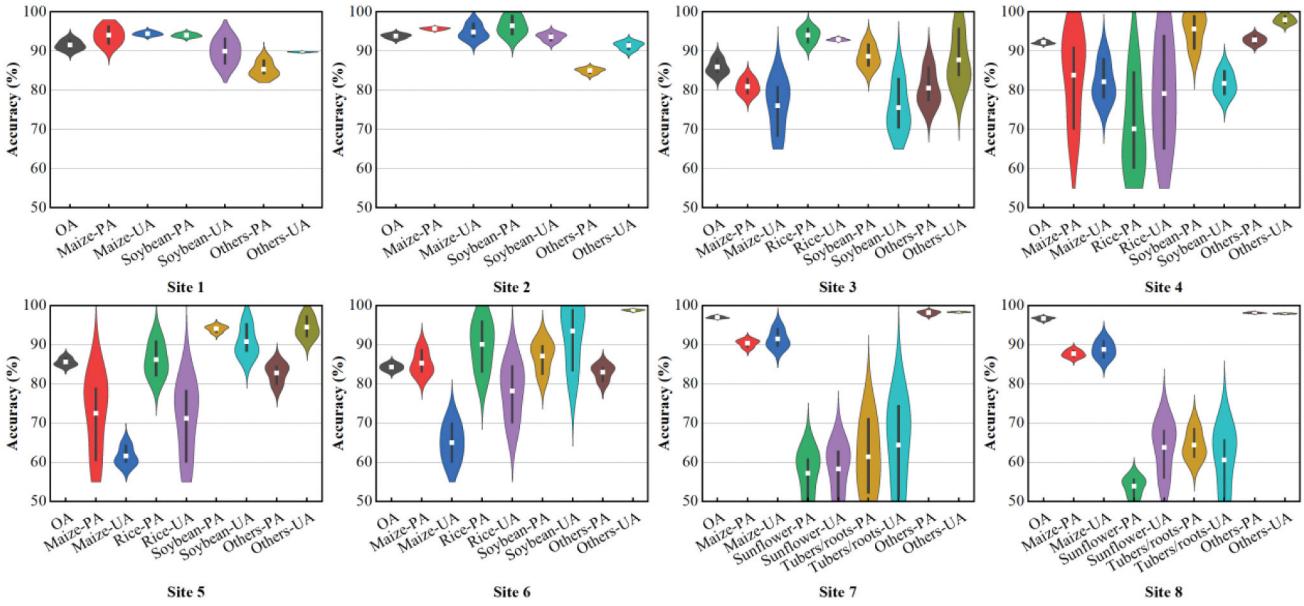

Figure 7. Accuracy evaluation results of temporal transferability at eight sites from 2020 to 2022 of the DSH model during the peak growth season, and the mark "□" denotes the average value.

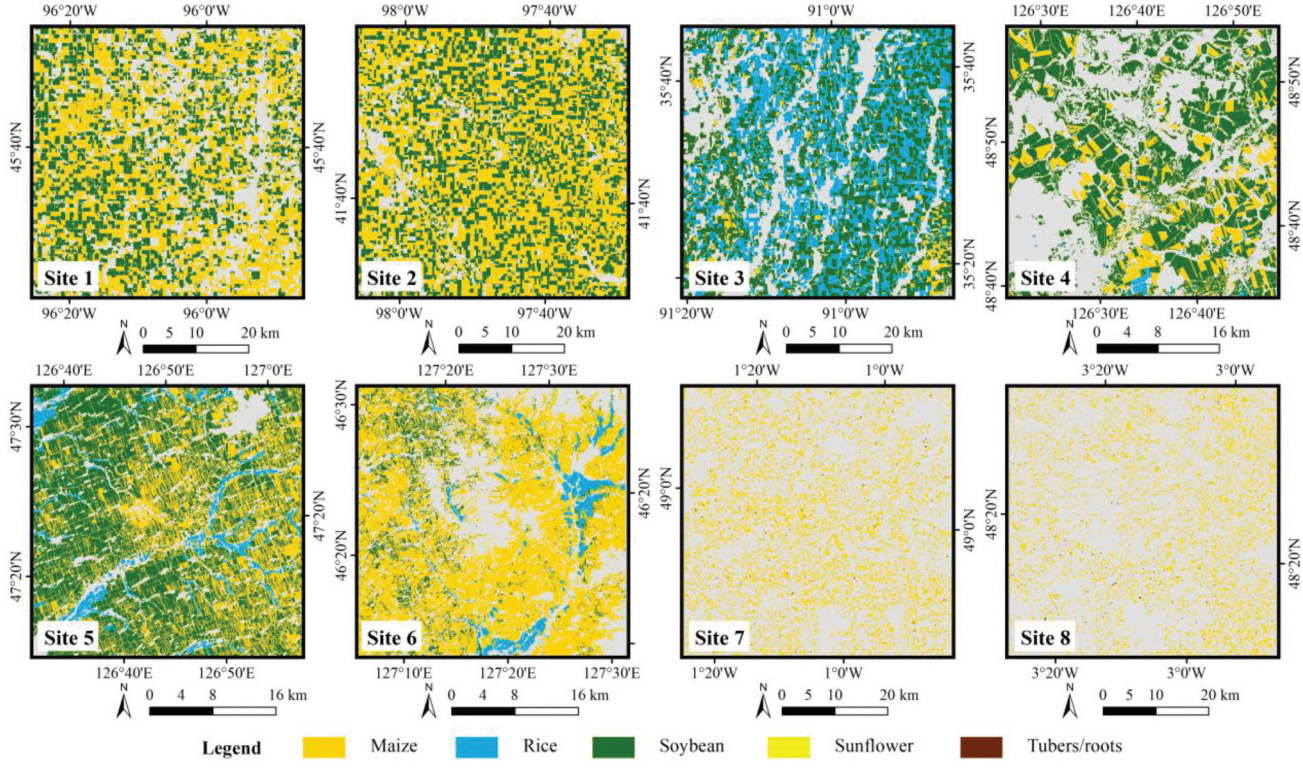

Figure 8. Crop classification results for eight study sites in the peak growth season of 2020.

4.2. Assessment of spatial transferability

We employed the model trained during the peak growth season of 2019 at site 1 and performed a full spatial transfer to sites 2–8 for the same year, with the assessment results presented in Table 2. Since the main crops at site 1 are maize and soybean, we assumed the model trained there was specialized in identifying these two crops. Consequently, at sites 2–6, we identified maize, soybean, and "others," while at sites 7–8, we identified only maize and "others"

Table 2. Accuracy evaluation results of full spatial transferability at sites 2–8 in 2019 of the DSH model trained at site 1 during the peak growth season.

	Maize		Soybean		Others		
Sites	PA(%)	UA(%)	PA(%)	UA(%)	PA(%)	UA(%)	OA(%)
Site 2	95.9	97.2	95.8	94.9	85.1	88.8	94.4
Site 3	84	81.2	84.5	83.8	83.9	82.2	83.7
Site 4	90.1	81.5	59.1	85.3	87.7	84.3	85.9
Site 5	71.1	68.3	78.4	79.6	79	80	75.6
Site 6	68.7	76.2	70.6	69.2	67.9	87.5	71.3
Site 7	85	83	—	—	89.3	91.8	90
Site 8	84.7	83.3	—	—	88.9	92.8	89.3

because soybean was not cultivated. Compared to the temporal transfer results of the model trained at the local site, the full spatial transfer exhibited decreased accuracy metrics except for sites 2–3. This is primarily attributed to the fact that sites 1–3 are located in the Great Plains of the United States, where large crop fields and well-defined borders prevail. Therefore, the model trained at site 1 showed robust spatial transfer performance at sites 2–3. However, sites 4–6 in northeast China and 7–8 in France showed a great decline in the full spatial transfer performance of the model trained in the Great Plain due to the fragmented plots and intricate planting structures.

Figure 9 further illustrates the spatial transfer performance of the model under similar planting structures. Compared to the full spatial transfer, except for site 3, which decreased by 2.1%, the accuracy of other sites was improved, especially site 6, which increased

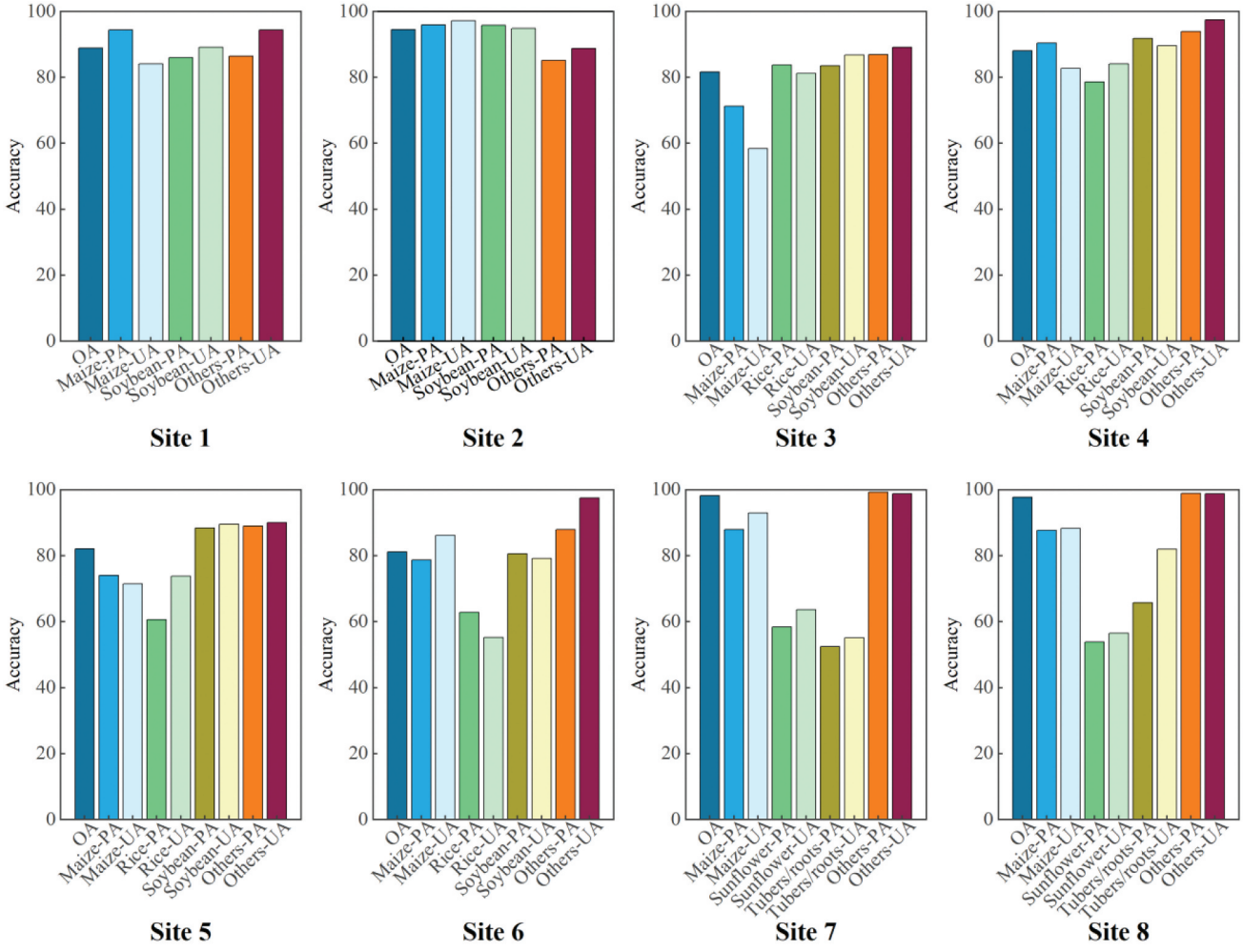

Figure 9. Accuracy evaluation results of similar planting structure spatial transferability during the peak growth season in 2019.

by 9.9%. The decrease in accuracy at site 3 under similar planting structures is due to site 3 being the result of the prediction of the model trained at sites 4–6. While sites 3–6 were all cultivated with maize, rice, and soybean, there is a significant difference in the planting proportions of these three crops at sites 3 compared to sites 4–6 (Figure 2). Models trained at sites 4–6 to identify maize and soybean may experience performance degradation when applied to site 3, where rice is the dominant crop. In general, compared with the full spatial transfer, the spatial transfer under similar planting structures showed a significant improvement in accuracy at 6 out of 7 sites. Therefore, for the spatiotemporal transfer experiment of the DSH model, we elected to conduct it between sites with similar planting structures.

4.3. Assessment of spatiotemporal transferability

The results of the spatiotemporal transferability assessment of the proposed DSH model are presented in Figure 10. Although the accuracy of spatiotemporal transferability decreased compared to the temporal transferability experiments, the DSH model still performed well with an average OA of 88.1% at eight sites in three years (2020–2022). It is noteworthy that sites 1 and 2 exhibited a minor decrease in average OA by 2.4% and 0.7%, respectively. Sites 3–6 experienced a more pronounced decline, with average OA reductions of 5%, 4.8%, 4.17%, and 4.73%, respectively. In contrast, sites 7 and 8 maintained consistent performance, with no significant change in average OA. This may be because of the similar planting proportions between sites 1 and 2, and sites 7 and 8. On the contrary, sites 3–6 had different planting proportions, with rice and soybeans being the dominant crops at site 3, while corn and soybeans were the dominant crops at sites 4–6. Additionally, due to different farming systems, the landscapes at sites 4–6 were relatively small and fragmented compared to site 3, which may lead to mixed pixels and thus affect accuracy. In summary, similar planting structures is crucial for spatiotemporal transferability.

4.4. Comparison with baseline models

Figure 11 shows the quantitative accuracy evaluation results of the four models (DSH, HRNetV2, DeepLabV3+, and RF) for the eight study sites in 2020. The proposed DSH yielded the best crop classification performance, attaining an average OA of 91.2%, outperforming by HRNetV2 (87.6%), DeepLabV3+ (86.6%), and RF (83%). The OA of sites 5 and 6 among the eight sites was lower than that of the other six sites.

As shown in Figure 12, F1 score was used to further measure the classification accuracy of each

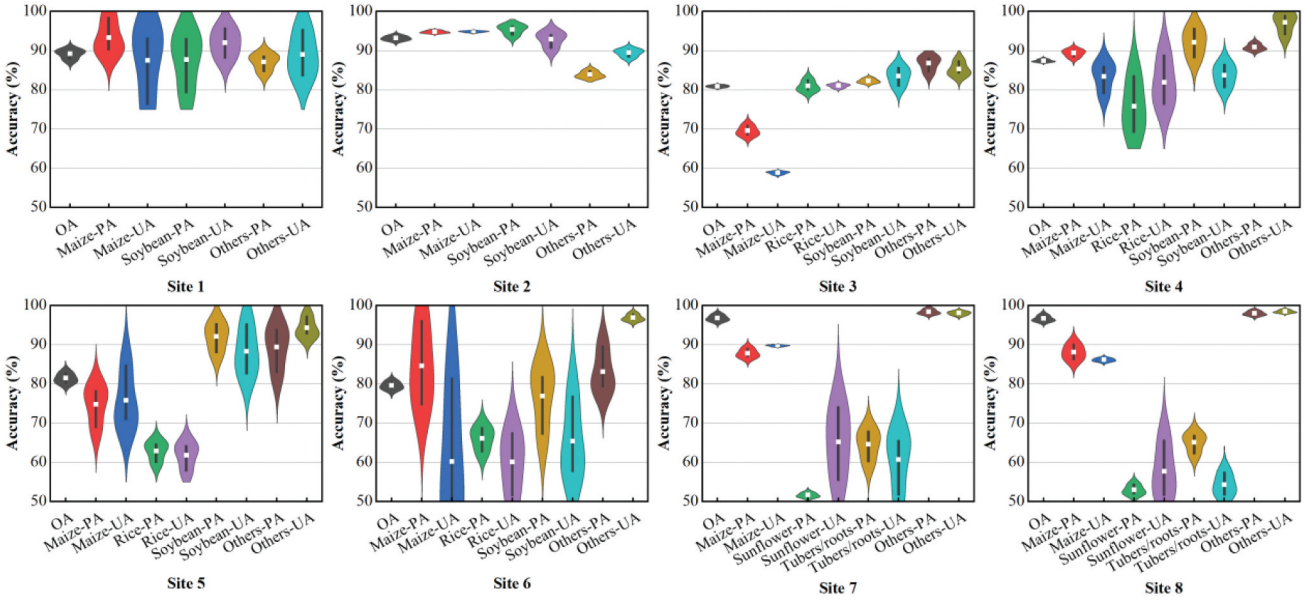

Figure 10. Accuracy evaluation results of spatiotemporal transferability at eight sites from 2019 to 2022 of the DSH model during the peak growth season, and the mark "□" denotes the average value.

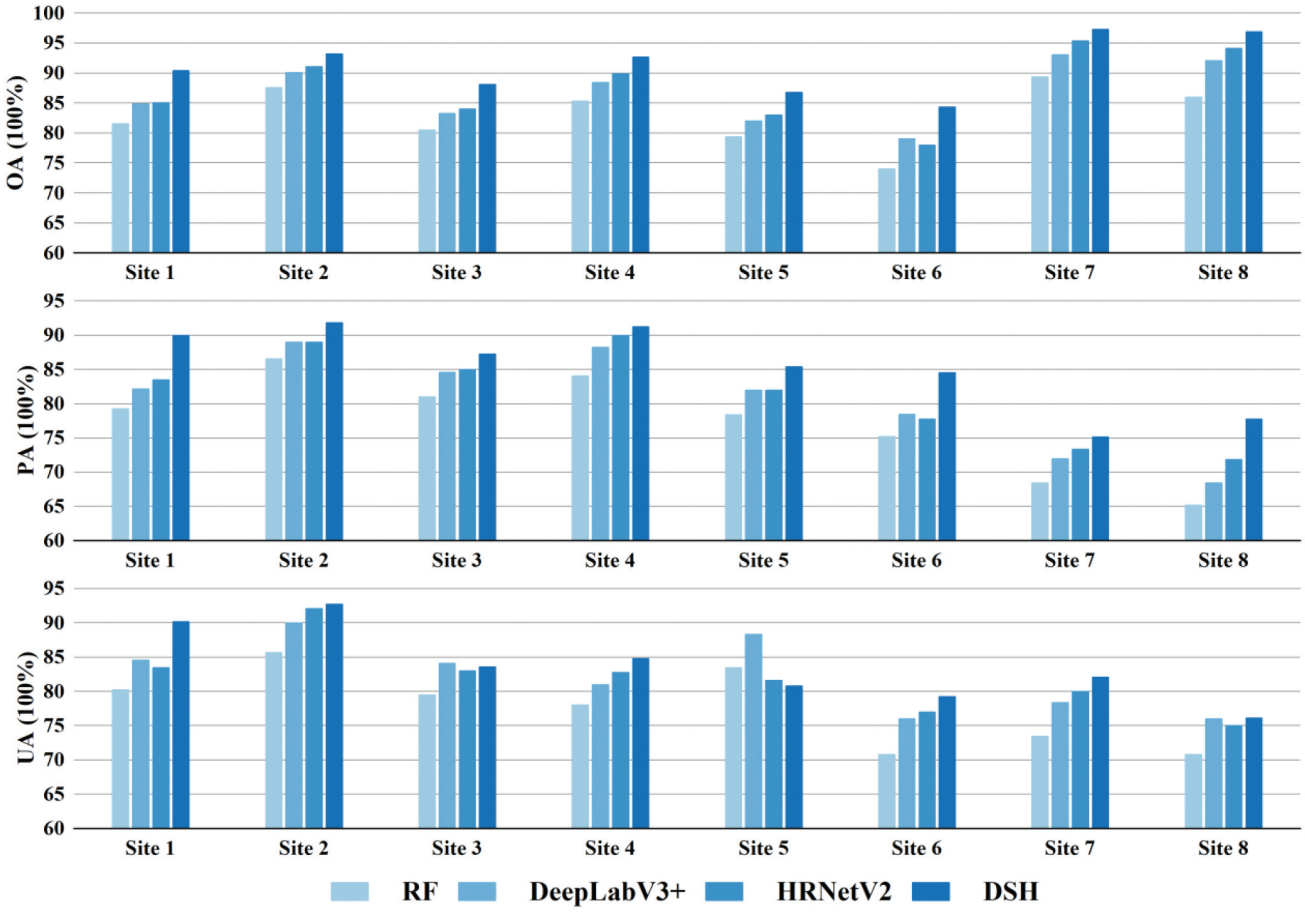

Figure 11. Accuracy evaluation of RF, DeepLabV3+, HRNetV2, and DSH.

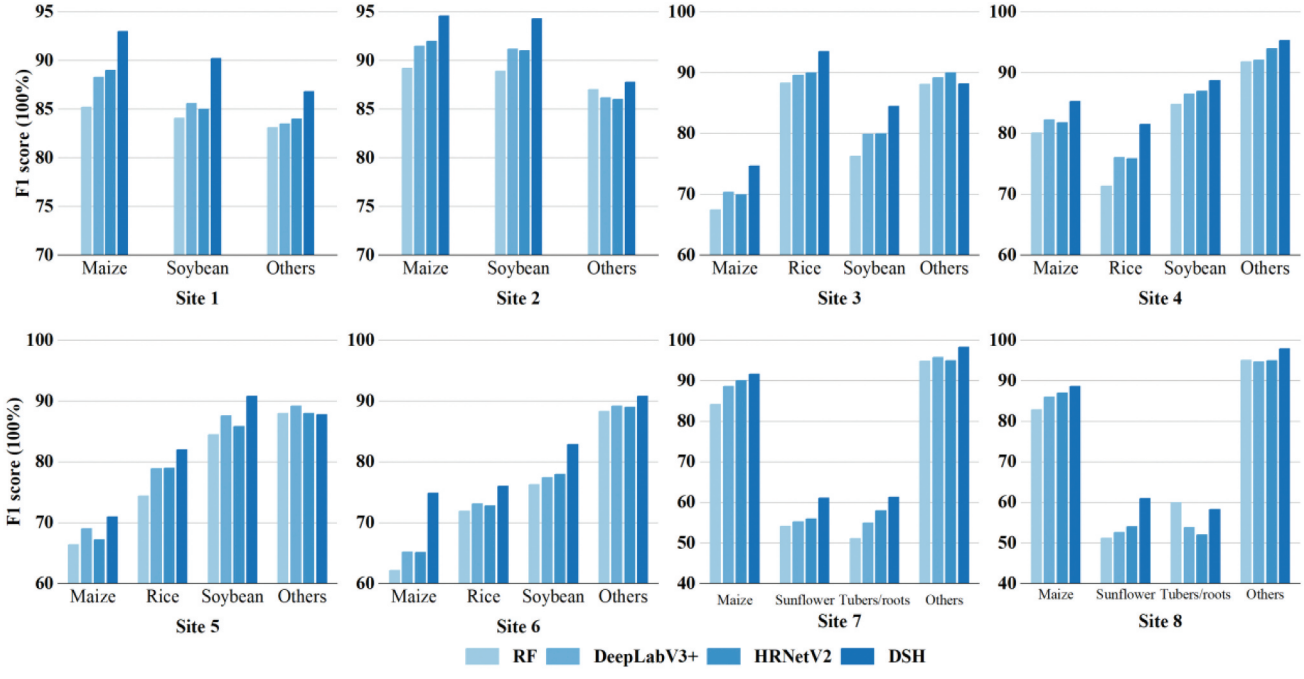

Figure 12. F1 scores of different land cover types classified by RF, DeepLabV3+, HRNetV2, and DSH.

land-cover type for the four models. For maize, rice, soybean, sunflower, and tubers/roots, DSH had higher classification accuracy than RF, DeepLabV3+, and HRNetV2, while for "others" types, its accuracy was similar to RF, DeepLabV3+, and HRNetV2. This indicates that the proposed DSH model has more advantages in distinguishing crop types. Sunflower showed lower accuracy than maize at sites 7–8, potentially because of the relatively low proportion of sunflower samples (less than 1%). The same trend was observed for maize at sites 3, 5, and 6, and tubers/roots at sites 7–8. In addition, it is worth noting that rice pixel at site 6 also had lowest proportion, but it had higher F1-score than maize, maybe due to the rice had large field parcels at site 6.

Figure 13 shows the classification results of the four models for the subregions of the eight sites. The proposed DSH method could more effectively distinguish crop types with smaller planting ratios than RF, DeepLabV3+, and HRNetV2, such as rice at sites 4–6 and tubers/roots at sites 7–8. Overall, the accuracy ranking of the three models was as follows: DSH > HRNetV2 > DeepLabV3+ > RF.

4.5. Comparison with full time series method

The temporal transfer accuracy comparison of DSH models trained using peak growth season (August) image and the full growth season (May to October) images monthly synthesized in 2019 at site 2 is presented in Table 3. The full time series methods exhibited slightly higher OA than the peak growth season method in 2020 and 2022, while the OA was slightly lower in 2021. For the DSH model established during the peak growth season, the average OA, maize's F1-score, and soybean's F1-score from 2020 to 2022 were 93.86%, 95.27%, and 94.97%, respectively. For the DSH model established using the full growth season monthly synthesized images, the average OA, maize's F1-score, and soybean's F1-score from 2020 to 2022 were 93.93%, 95.37%, and 95.03%, respectively. Overall, the full time series and peak growth season methods achieved comparable performance, but the former used six times more data than the latter. Obtaining time series data during full growth season is a time-consuming and labor-intensive task, especially for large-scale crop mapping. In addition, the peak growth season method can provide classification results two months before harvest, highlighting the advantage of the proposed DSH method in early crop mapping.

5. Discussion

5.1. Contribution of histogram matching to improving the transferability of DSH

Figure 14 shows the histograms of the ten bands for the reference training image, raw test images, and test images after histogram matching. The spectral value distribution of the raw test images (Figure 14, (b,d,f)) differed significantly from that of the reference training image (Figure 14, (a)), whereas the histogram of the matched test images (Figure 14, (c,e,g)) were similar to that of the reference training image (Figure 14, (a)). In particular, the gray-scale distribution of the raw test image in 2022 showed a considerable difference from that of the raw test images in 2020 and 2021. The reason for this difference was the variation of the range of spectral values. According to the Copernicus Data Center, after 25 January 2022, S2_SR images had their DN value range shifted by 1000 (Copernicus 2021). For example, in 2020, the RE1 curve was represented as a single peak in the raw test image, while it showed two peaks in both the reference training image and the matched test image. Conversely, the SWIR1 curve showed two peaks in the raw test image and a single peak in the reference training image and matched test image.

To estimate the effects of histogram matching, comparison experiments were conducted to evaluate the classification accuracy between the raw test images and the histogram matched test images during the peak growth season. The results are shown in Figure 15. The matched test images demonstrated stable performance over three years, with average OA, PA, and UA values of 90.9%, 84.8%, and 84%, respectively, across the eight study sites. The raw test images demonstrated comparable performance to the matched test images in 2020 and 2021 but showed a decrease in performance in 2022. In 2022, the average OA, PA, and UA of the raw test image at eight sites were 70.8%, 62.5%, and 59.4%, respectively, with OA values at sites 1–8 falling below 80%. This decrease is attributed to the significant change in the spectral value range of the raw test images in

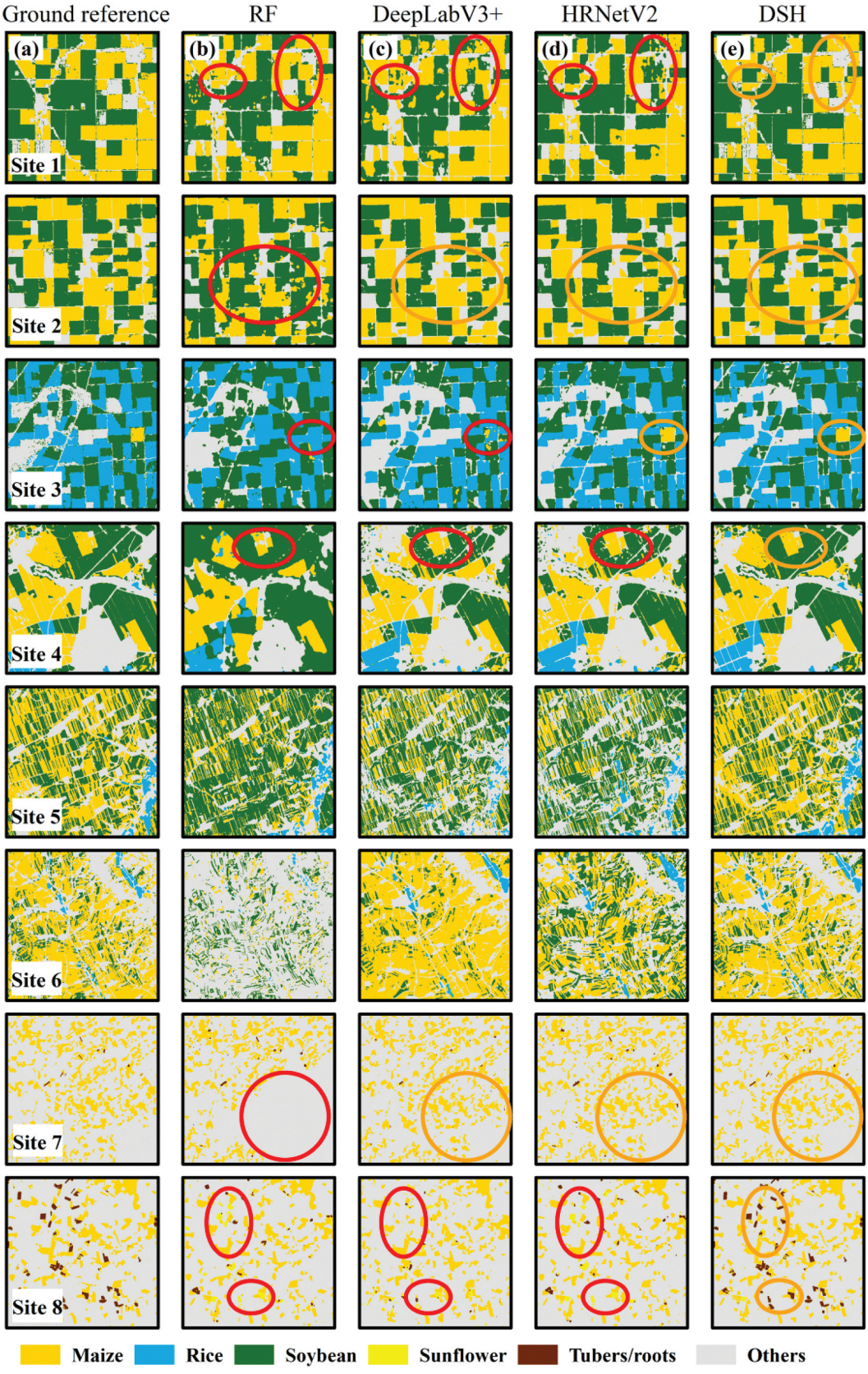

Figure 13. Details of the crops map obtained for each study site with different models in 2020. (a) Ground reference, (b) classification results of RF, (c) classification results of DeepLabV3+, (d) classification results of HRNetV2, and (e) classification results of DSH. The red ellipse box indicates the error classification details, and the orange ellipse box indicates the correct classification details.

Table 3. Accuracy comparison among full time series and the peak growth season methods at site 2 from 2020 to 2022.

		2020		2021		2022	
Classes	Accuracy indices (%)	Full season	Peak season	Full season	Peak season	Full season	Peak season
	OA	93.4	93.2	93.5	94	94.9	94.4
Maize	PA	95.7	95.8	95.1	96	97.6	95.4
	UA	93.3	93.5	94.5	94	95.5	97.1
	F1	94.5	94.6	94.8	95	96.5	96.2
Soybean	PA	93.5	94.1	95.3	96	97.4	99.1
	UA	94.5	94.4	93.9	94	95.6	92.5
	F1	94	94.2	94.6	95	96.5	95.7
Others	PA	86.4	85.5	86.5	85.3	84.7	84.2
	UA	90.1	90.3	90.5	92.2	92.3	91.8
	F1	88.2	87.8	88.5	88.6	88.3	87.8

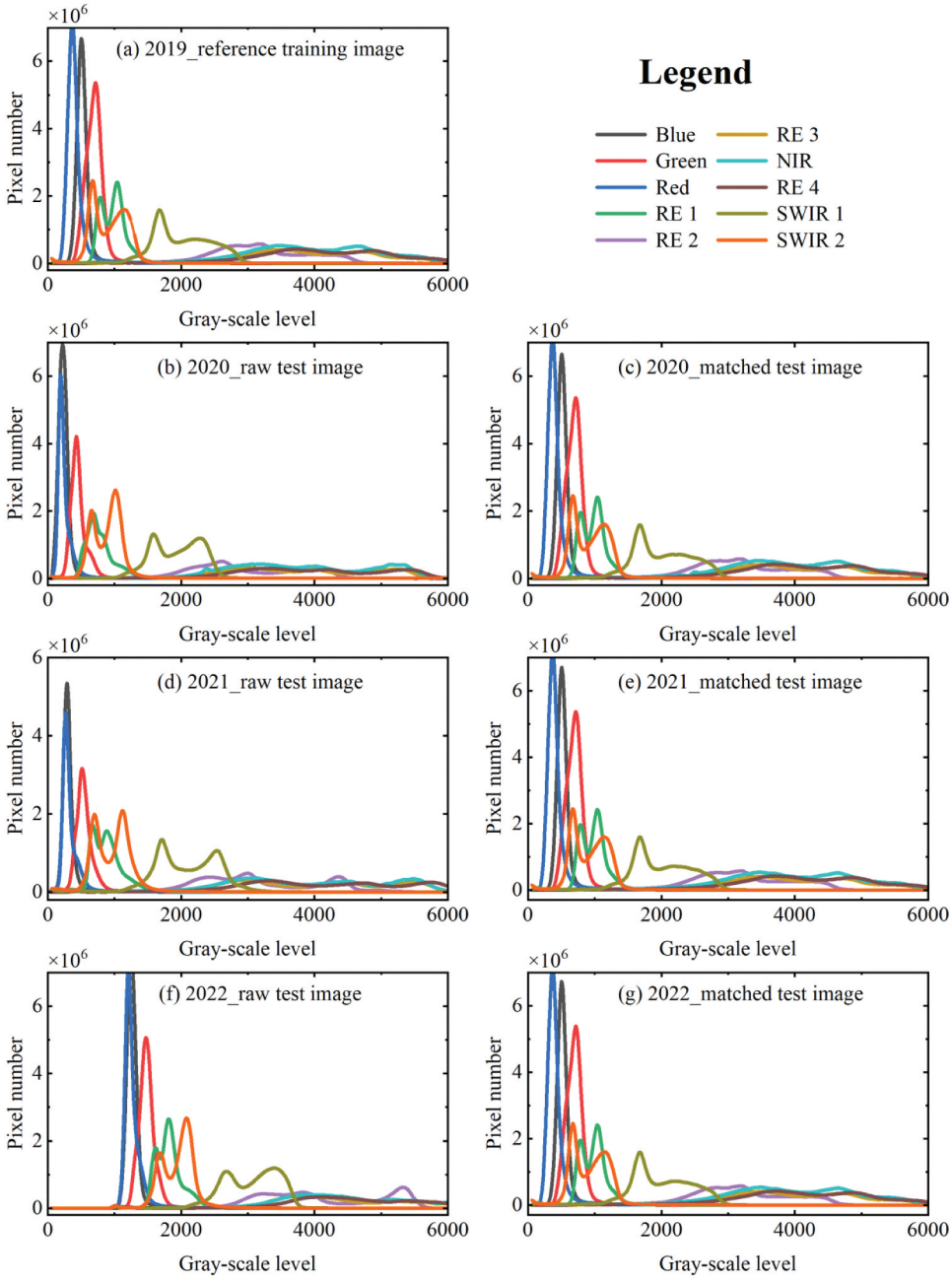

Figure 14. Histogram of reference training image from 2019 (a), raw test images from 2020-2022 ((b), (d), (f)), and histogram matched test images from 2020-2022 ((c), (e), (g)) at site 4.

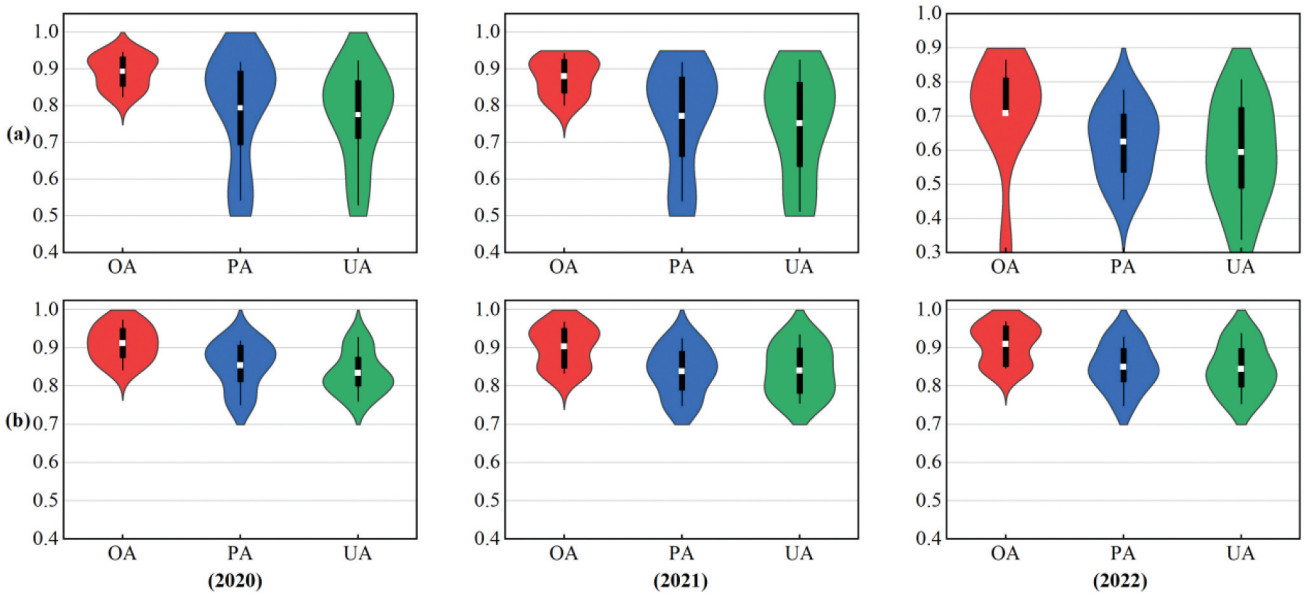

Figure 15. Accuracy evaluation at eight sites from 2020 to 2022 of the DSH model (a) for raw images and (b) for histogram matched images. The PA and UA denote the average PA and average UA of different land cover types, and the mark "□" denotes average value.

2022 (Figure 14). Figure 16 presents a subset from site 4 as an example, demonstrating that test images after histogram matching can more effectively identify the three crops. The corn classification result based on the raw test image had some omissions in 2020, and the rice classification result based on the raw test image had some omissions in 2021. Moreover, there were many errors in the boundaries of the three crop fields obtained from the raw test image in 2022. However, test images processed by histogram matching could accurately identify the crops from 2020 to 2022.

5.2. Impact of embedding position of channel self-attention module on classification

Taking the 2021 crop classification at sites 1–2 as an example, we explored how different embedding positions of the channel self-attention modules affect the performance of the DSH model. The specific embedding location is shown in Figure 4. Table 4 presents the results of the accuracy evaluation of the experiment. Compared to the original DeepLabV3+, accuracy was improved with all tested embedding positions at sites 1–2.

The loss curves for the original DeepLabV3+ and three DSH networks, each with a channel self-attention module, are presented in Figure 17. Compared to the original DeepLabV3+, the DSH models with channel self-attention modules exhibited smoother loss curves. Notably, DSH_P2 achieved faster convergence and was more easily optimized, particularly in the early stages of training. In DSH_P2, embedding the channel self-attention module within the high-level features of DeepLabV3+ allowed for the efficient weighting of numerous features, reducing computational load and enhancing network training efficiency. Although DSH_P1 and DSH_P2 showed similar convergence speeds, DSH_P2 consistently displayed lower loss values under identical hyperparameters. The four loss curves at sites 1–2 all followed a similar trend. Consequently, in this study, we chose to embed the channel self-attention module at high-level feature positions and integrate it into the decoder (P2).

5.3. Channel self-attention feature representations for crops

We examined the channel self-attention and high-level feature representations of the proposed DSH model to better understand its crop mapping capabilities, using site 4 as an example (see Figure 18). The channel self-attention mechanism captured the inter-channel correlations between pixels by calculating the channel weights for each hidden feature within the DeepLabV3+'s high-level feature maps. Overall, the self-attention feature maps (Figure 18, (c1)–(c3)) exhibited a more concentrated attention, primarily

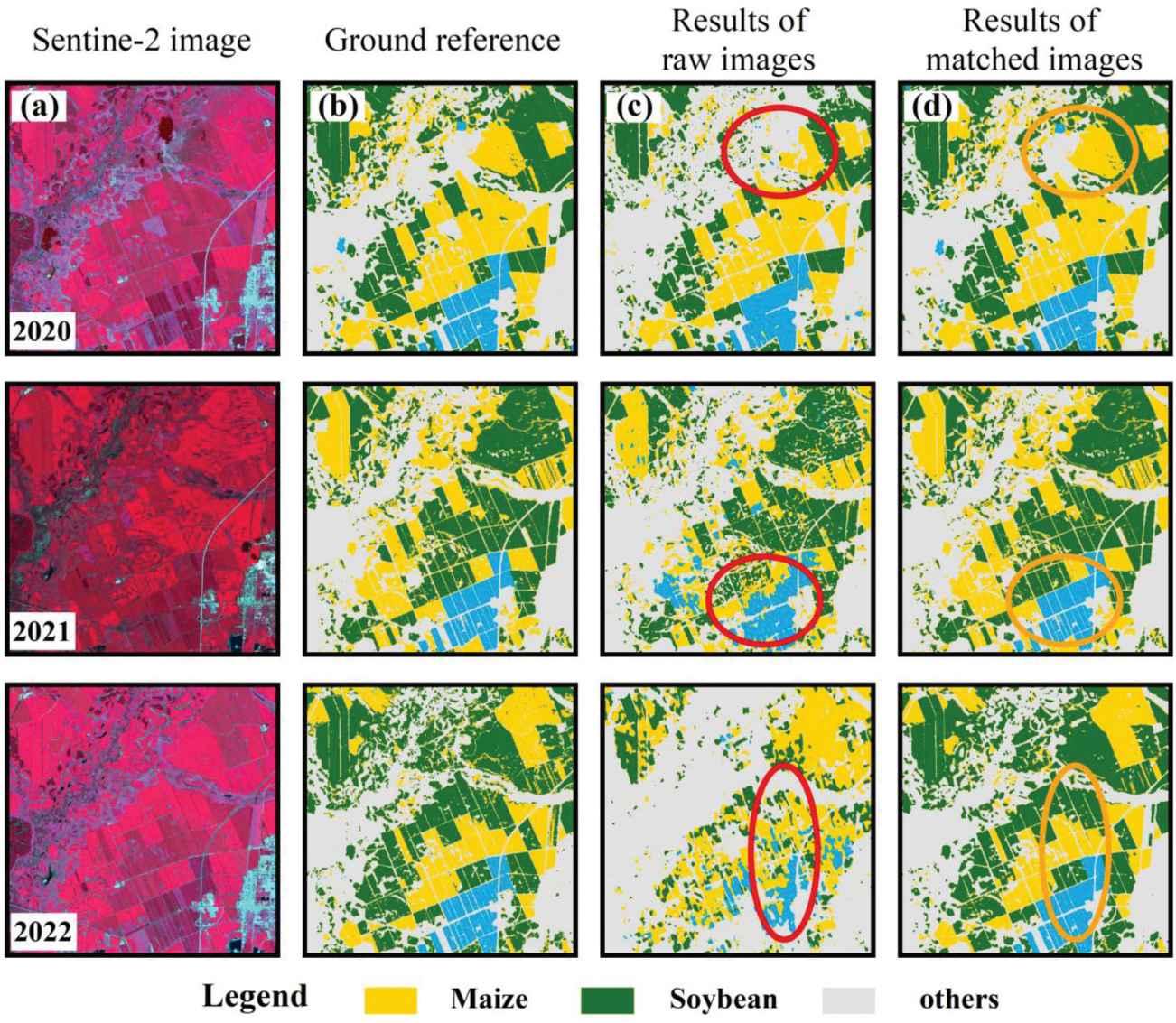

Figure 16. Detailed crop mapping results for raw and matched test images at site 4 from 2020–2022. (a) False-color composited sentinel-2 images during the peak growth period (R: NIR, G: red, and B: green), (b) ground reference, (c) classification results of raw images, and (d) classification results of matched images. The red ellipse box indicates the error classification details, and the orange ellipse box indicates the correct classification details.

Table 4. Accuracy comparison among DSH_P1, DSH_P2, DSH_P3 and the original DeepLabV3+ at sites 1–2 of 2021.

			Maize		Soybean		Others	
Sites	Embedded position	OA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
Site 1	DSH_P1	90	93.2	92.5	93	91	84.4	89
	DSH_P2	91.3	94	94	93.6	90	84.1	89.8
	DSH_P3	89.8	92.9	92.4	92.4	90	84.7	88
	DeepLabV3+	88.7	91	90.4	90.1	88.6	84.5	88.9
Site 2	DSH_P1	93.7	95.7	94.5	95.9	93.5	85.4	92
	DSH_P2	94	96	94	96	94	85.3	92.2
	DSH_P3	93.4	95.2	94.3	95.7	93	85.4	91.9
	DeepLabV3+	92.1	93.1	88.7	92.8	89.1	89.8	84.2

focusing on the internal class characteristics of the target crops rather than boundary information, which allowed for a more precise mapping of the target crops. Conversely, the attention of the DeepLabV3+ high-level feature maps (Figure 18, (d1)–(d3)) was relatively dispersed, with a broad scope of focus that encompassed crops but also included other types of land cover. This dispersed

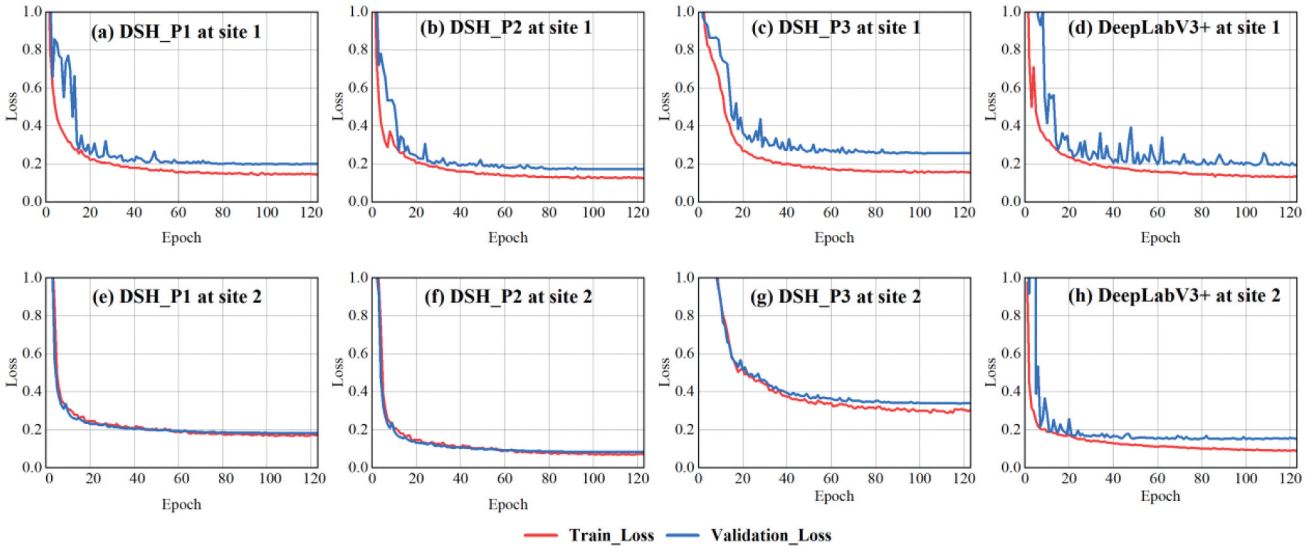

Figure 17. Loss curves of the different channel self-attention module embedding positions of DSH and DeepLabV3+ models during the training process at sites 1–2.

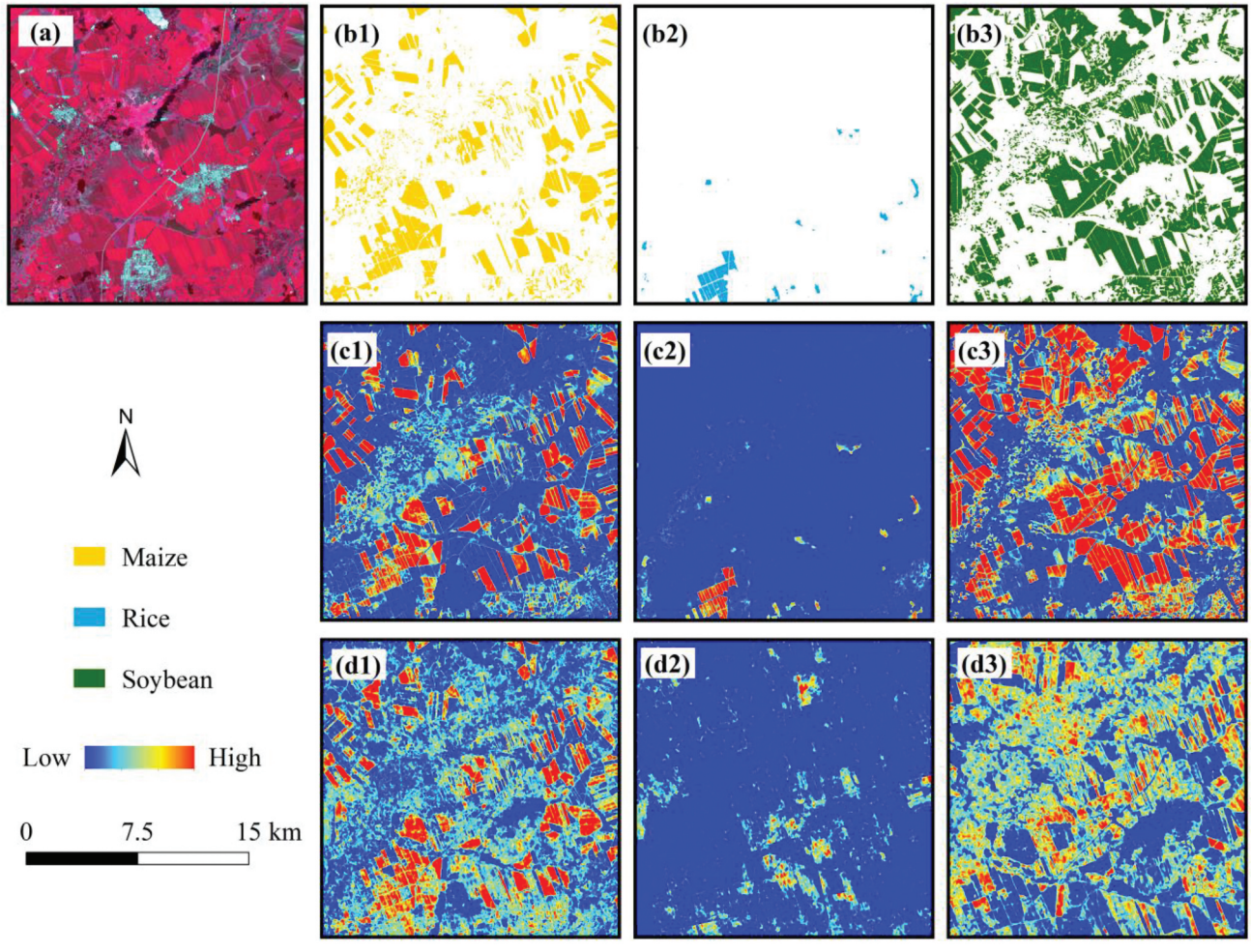

Figure 18. Learned channel self-attention and high-level feature maps: (a) false-color composited sentinel-2 images during the peak growth period (R: NIR, G: red, and B: Green); (b1), (b2), (b3) classification results of maize, rice and soybean; (c1), (c2), (c3) channel self-attention feature maps of maize, rice and soybean; (d1), (d2), (d3) original DeepLabV3+ high-level feature maps of maize, rice and soybean.

attention reduced the ability of the DeepLabV3+ high-level features to capture key information about the target crops. This suggested that the channel self-attention module, when integrated into the DeepLabV3+ high-level feature layers, both inherited and amplified the discriminatory power of DeepLabV3+'s features through the modeling of global relationships.

For instance, in the feature map of maize, the self-attention feature maps closely attended to maize fields (Figure 18, (c1)), whereas the DeepLabV3+ high-level feature maps were somewhat distracted by natural vegetation areas with similar morphologies (Figure 18, (d1)). In rice mapping, the self-attention mechanism successfully captured the orderly arrangement and spatial distribution characteristics unique to rice fields (Figure 18, (c2)), which the DeepLabV3+ high-level feature maps did not distinguish with the same precision (Figure 18, (d2)). Similarly, in the soybean mapping task, the self-attention mechanism maintained a high-level of attention to both regular and fragmented soybean fields (Figure 18, (c3)), while the DeepLabV3+ high-level feature maps nearly covered all land cover categories (Figure 18, (d3)). Essentially, the channel self-attention module acted like an efficient internal selector, prioritizing the most discriminative channels that carry the key information for differentiating crops such as maize, rice, and soybean. This attention-based improvement significantly enhanced the accuracy of classification tasks, enabling the enhanced feature maps to more clearly delineate the crops of interest and providing a more reliable foundation for subsequent crop mapping.

5.4. Future work

DSH can enhance the weight of important features and reduce the domain gap between the training image and the test image, thereby improving the model's generalization performance under phenological period differences. However, it still has some limitations.

First, the growth period of crops vary across different months, leading to significant changes in the range of image values from month to month. A model established in a certain month cannot be used to generate crop map in other months. Therefore, we trained the DSH model for each month of the crop growth season, which means there is one DSH model per month. Our validation results indicate that applying the model trained in a certain month of 2019 to the corresponding months from 2020 to 2022 can ensure accurate crop identification. However, when input images are heavily polluted by clouds and shadows, and even missing, DSH might be ineffective. One possible future work is to investigate the applicability of the combination of data fusion techniques and the DSH model. Moreover, the development of transfer methods between months is also a direction for the future.

Second, our experimental results show that DSH exhibits higher accuracy and transferability in areas with similar planting structures and regular land parcels (such as sites 1 and 2, and sites 7 and 8), while its spatial transferability decreases in areas with different planting structures. Based on these findings, the direct application of the DSH model in regions with non-traditional agricultural practices or underrepresented crops in the training data (such as South America or Southeast Asia) might constrain its performance. This is potentially due to fragmented land parcels or multiple cropping per annual. Therefore, for future promotion and application of DSH, the entire research area can be partitioned according to certain rules, and the DSH model established within a specific sub-area is expected to extend effectively to other similar sub-areas. Additionally, for areas with fragmented land parcels, it is necessary to further enhance the stability of the DSH algorithm.

Finally, to ensure an adequate sample size, we utilized the CDLs data products from eight sites directly, without implementing a further selection process. Although the DL methods have a certain tolerance for noisy samples, it is evident that high-quality samples can further enhance the classification accuracy of the model (Zhang et al. 2016). Therefore, in future studies employing CDL products as training samples, for those CDL products with associated confidence levels, such as CDL data or Theia's land cover product, we intend to adopt the strategy outlined by reference (Hao et al. 2020). This approach involves the selection of high-confidence pixels to serve as training samples, thereby enhancing the robustness of our models. For products lacking confidence scores, it may be necessary to generate high-confidence training samples through some confidence learning strategies (Northcutt, Jiang, and Chuang 2021).

6. Conclusion

A spatiotemporally generalizable DL approach, DSH, was developed based on the DeepLabV3+ architecture with self-attention mechanisms and histogram matching. The temporal transfer experiment results showed that the DSH trained during the peak growth season had the highest temporal transferability, and the average OA of DSH in 2020–2022 were 90.9% at the eight sites. The results of spatial transfer experiments showed that the spatial transfer under similar planting structures had higher accuracy than that of full spatial transfer. Consistent with this observation, spatiotemporal transferability analyses confirmed an average OA of 88.1% for DSH at the eight sites from 2020 to 2022. These results underscored the critical role of similar planting structures in facilitating effective spatiotemporal transfer. Compared with RF, DeepLabV3+, and HRNetV2, the proposed DSH demonstrated the most successful crop classification performance at all eight study sites, with an average OA of 91.2%, which was superior to RF (83%), DeepLabV3+ (86.6%), and HRNetV2 (87.6%) in 2020. DSH during peak growth and full growth season achieved comparable performance (average OA = 93.9%). The accuracy of image classification after histogram matching remains stable from 2020 to 2022, while the accuracy of image classification without histogram matching declines significantly in 2022. This indicates that histogram matching contributes to the improvement in the transferability of DSH. The embedding of a channel self-attention module into the position of high-level features is crucial for improving the DSH performance. Visualizing features from the channel self-attention and high-level layers of DeepLabV3+ revealed that integrating the channel self-attention module with DeepLabV3+'s high-level features significantly improved crop mapping by enhancing relevant semantic information. Thus, the DSH has high spatiotemporal generalizability and low input data requirements, making it suitable for large-scale crop mapping.

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Disclosure statement

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Data availability statement

The Sentinel-2 Level-2A surface reflectance (SR) data used in this study can be accessed through the Google Earth Engine cloud computing platform (https://earthengine.google.com/). The United States Department of Agriculture (USDA) Cropland Data Layer (CDL) were available at CropScape: A Web service-based application for exploring and disseminating US conterminous geospatial cropland data products for decision support, Computers and Electronics in Agriculture (https://nassgeodata.gmu.edu/CropScape/). Theia's land cover product can be obtained through (https://www.theia-land.fr/en/product/land-cover-map).

Available code

https://github.com/Maria-hyChe/DSH.

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